Infectious Bursal Disease (GUMBORO DISEASE)
傳染性華氏囊病

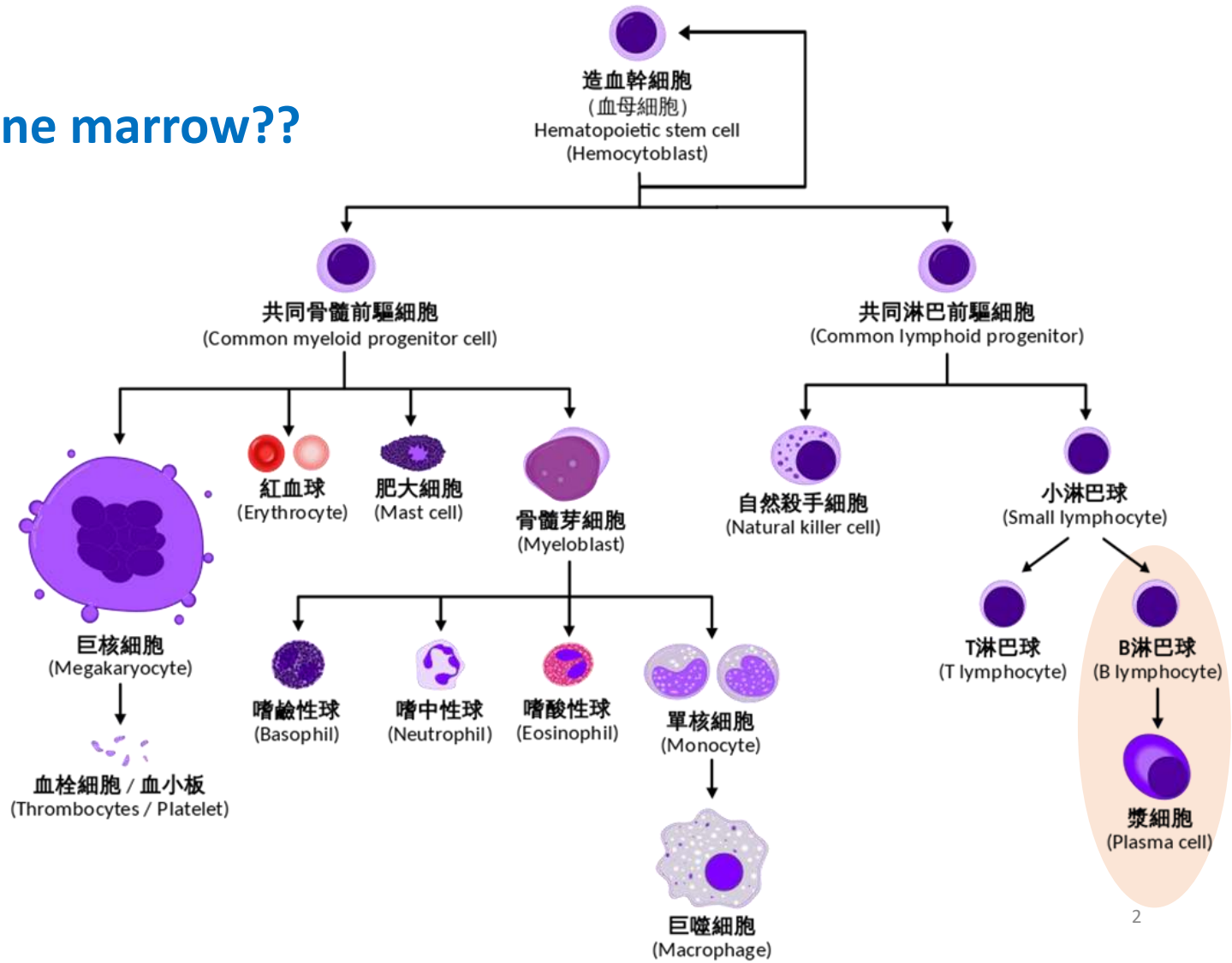
Nai-Huei Wu, DVM, PhD

Lymphocytes

What is Lymphocyte?

T cells - Thymus

B cells = From ?? Bone marrow??



免疫系統的結構和功能

Lymphoid organs

Primary:

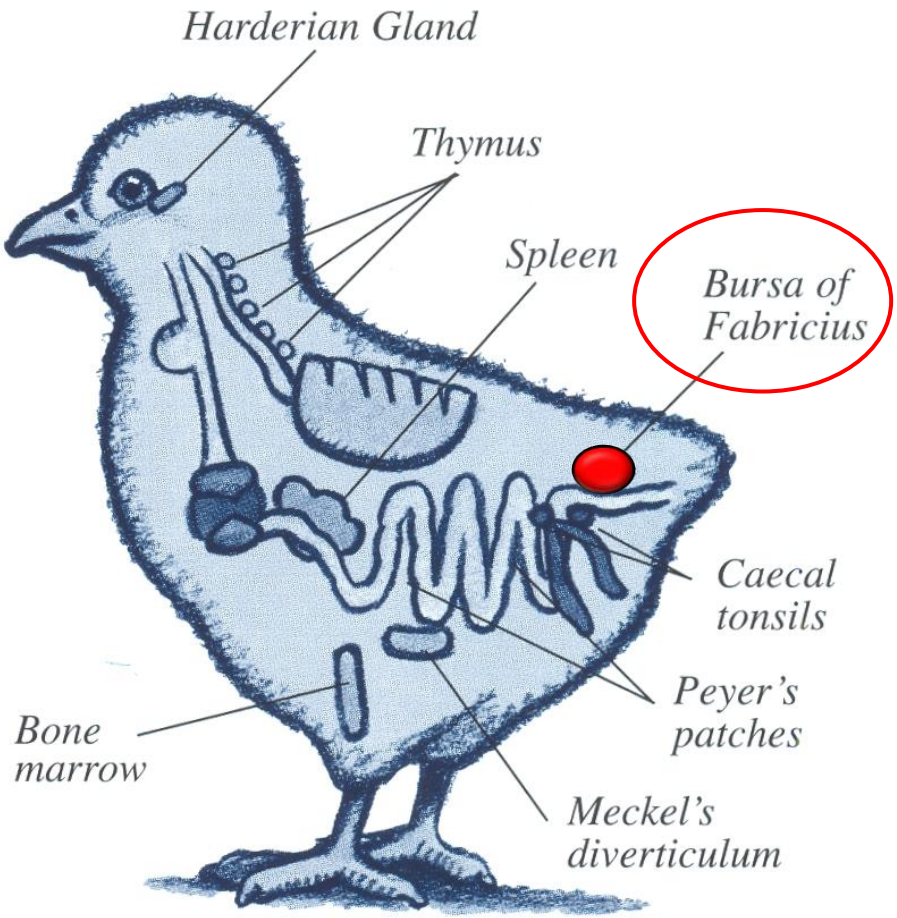
- Thymus (T lymphocytes)
- **Bursa of Fabricius** (B lymphocytes)

Secondary:

- Spleen
- Harderian gland 哈氏腺
- Bone marrow

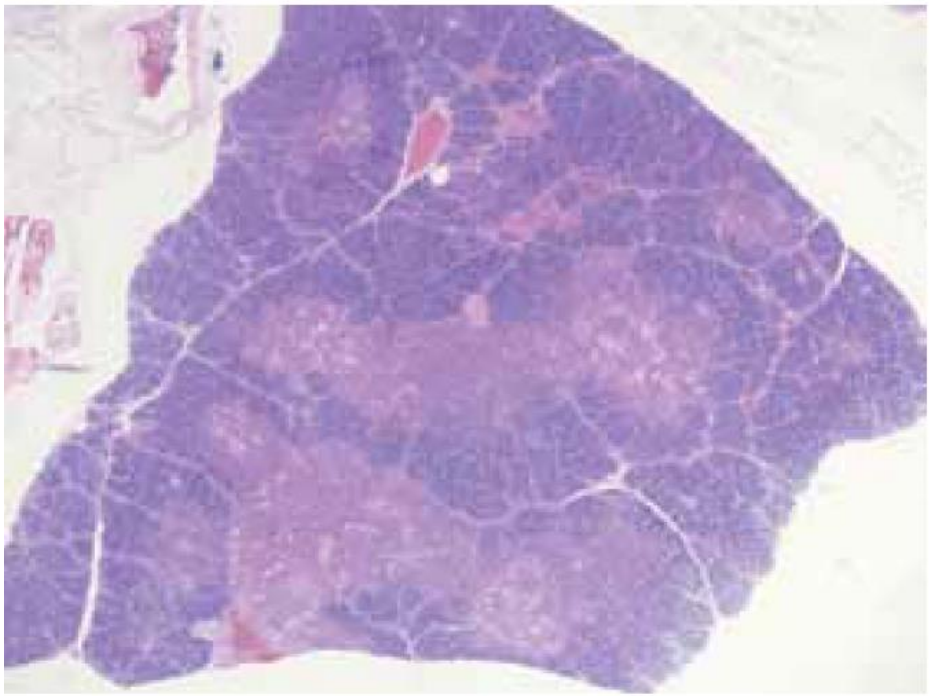
Sub-lymphoid tissue

- Conjunctival (CALT)
- Intestinal (GALT)
- Bronchial (BALT)

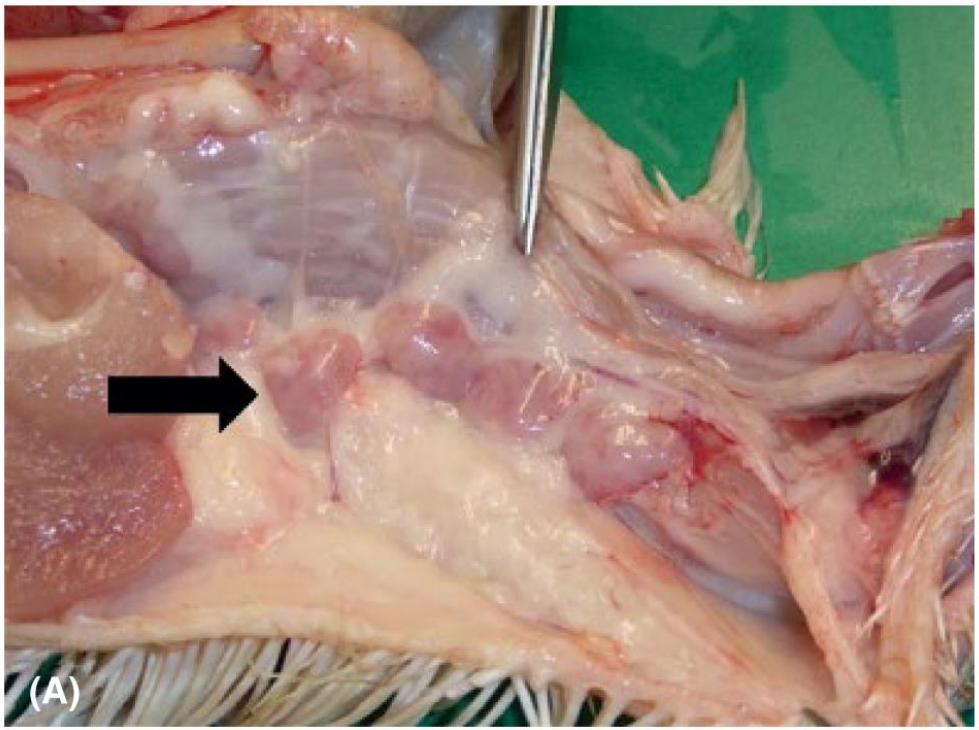


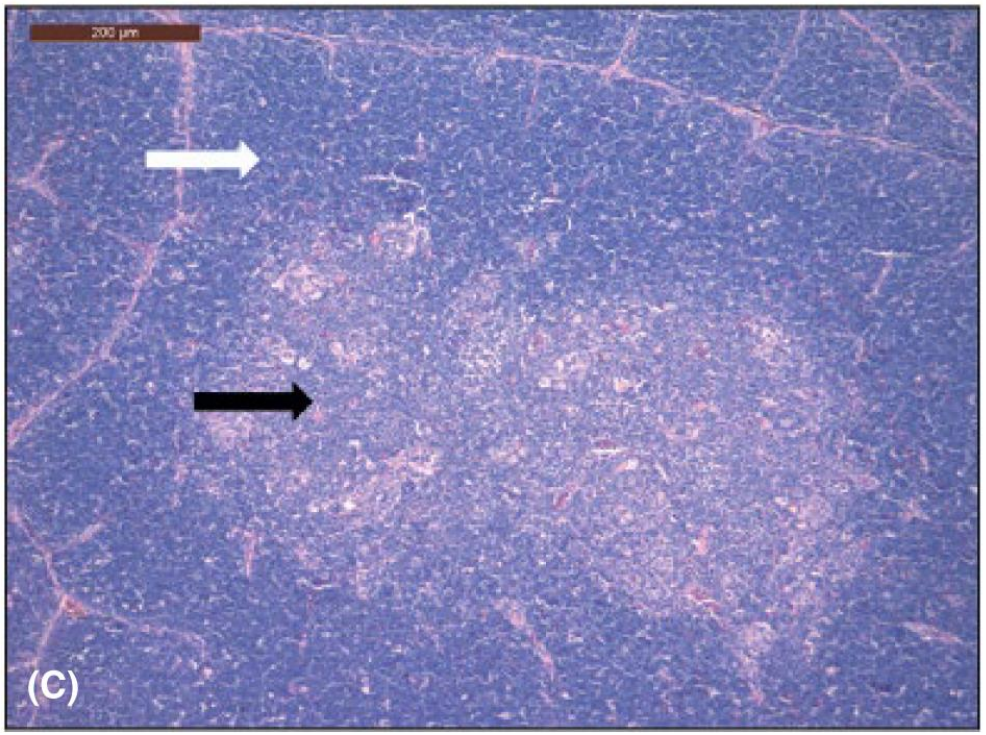
Thymus



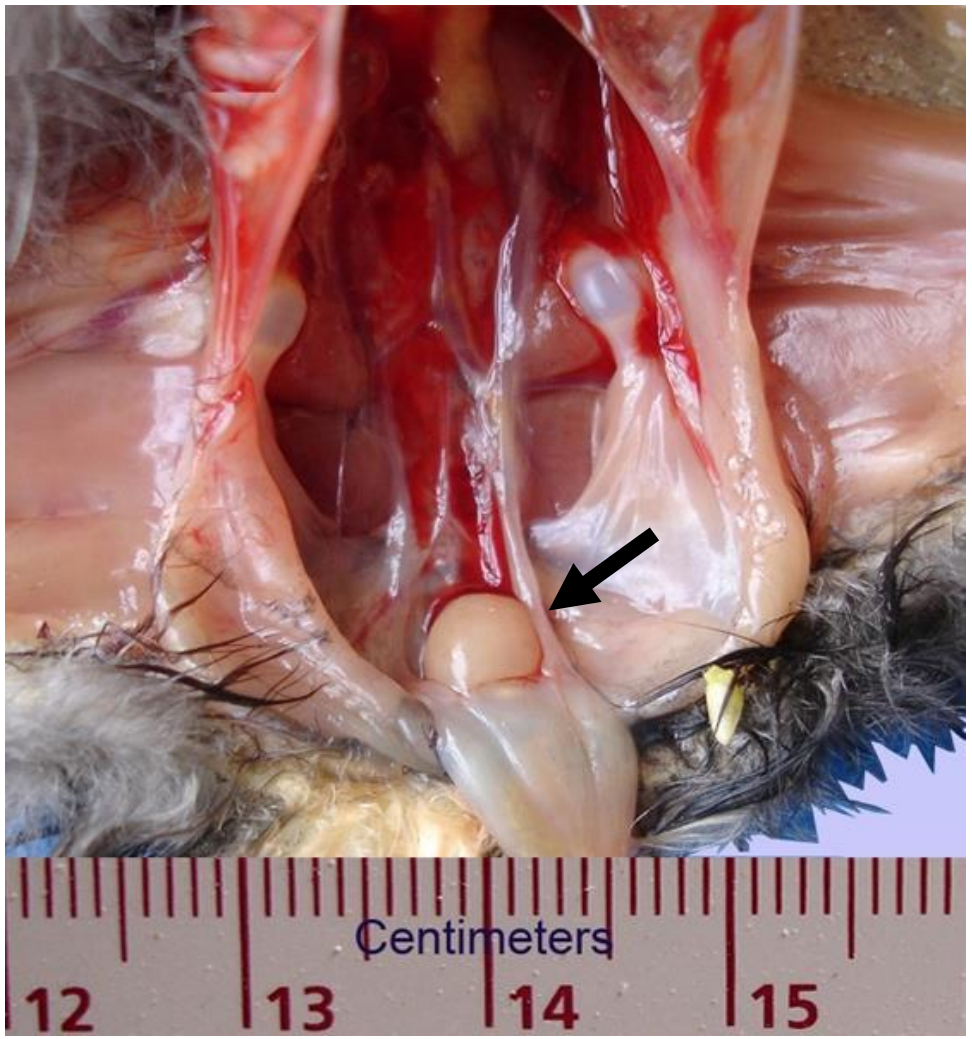


Thymus

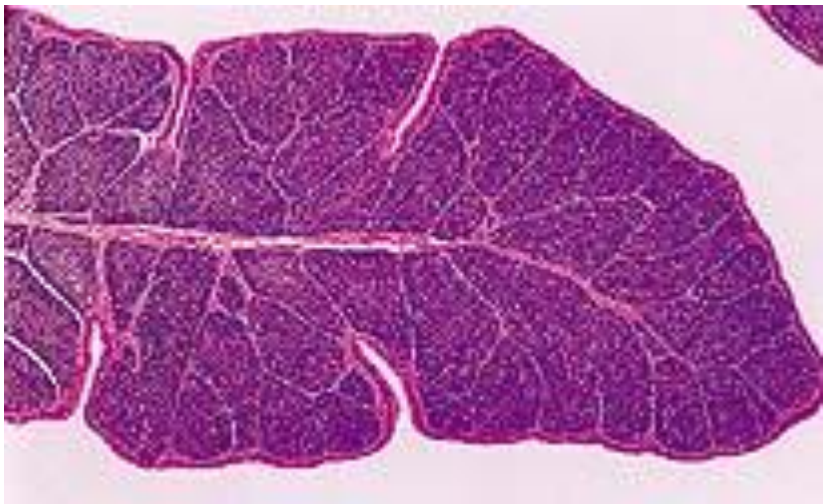




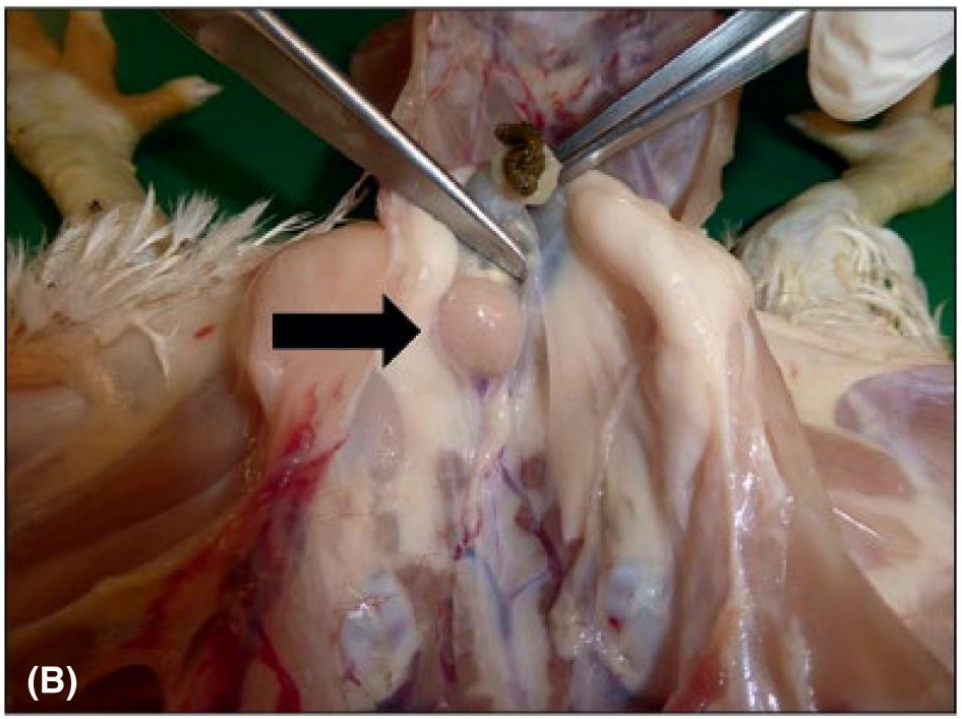
Bursa of Fabricius

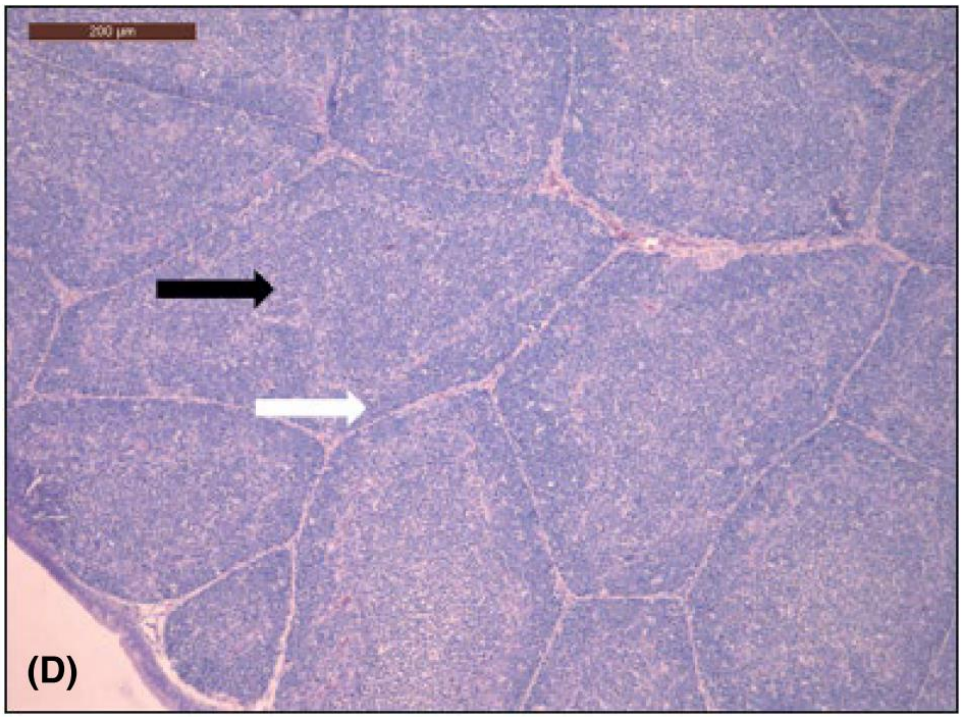






Bursa of Fabricius





Cecal tonsils





Bursa of Fabricius

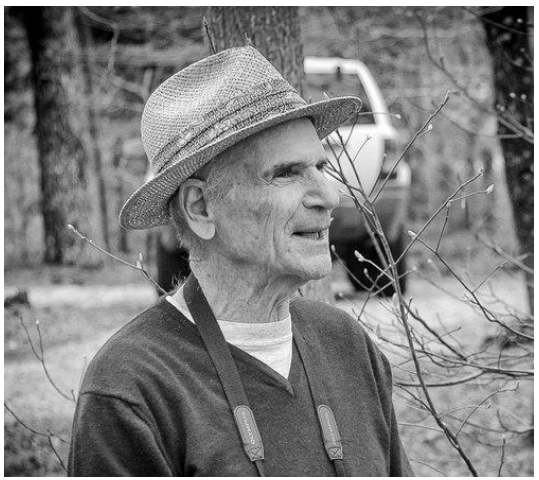


- Hieronymus Fabricius

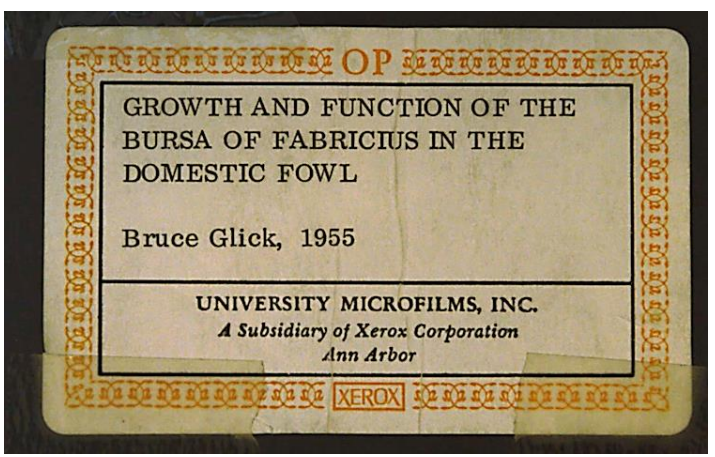
  Italian-(Girolamo Fabrizio)

- 1533-1619
- 解剖學家
- 1621年發表的文章

  第一次完整描述 Bursa的構造



- Bruce Glick
- 1956發現移除Bursa→
  - **抗體**無法形成



Adaptive or acquired immunity in chicks

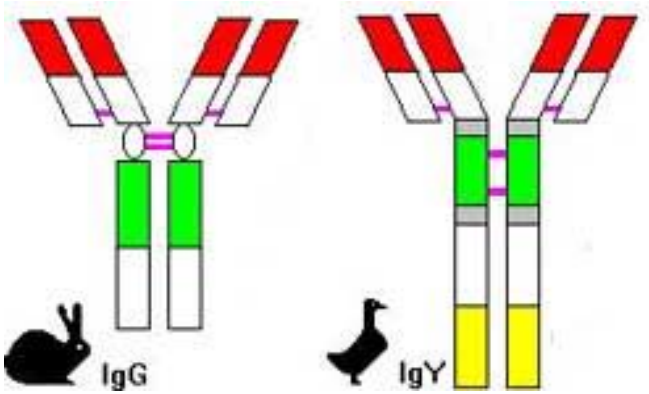
Active immunity

- specific immune response against antigens
  - Primary response (1st contact with antigen)
    - IgM → **IgY**
    - Immune memory (**B cells**)
  - Secondary response (2nd contact with antigen)
    - Rapid and longer immune response
    - **IgY**

Passive immunity

- Maternal antibody (transmitted from hen to chick via egg yolk)
- Transfusion of antibodies by sera

Humoral immune response



Three Ig classes: IgM, IgY, IgA

- IgM: pentamer, expressed on the surface of B cells
- IgY: functionally homologous to mammalian IgG
- IgA: dimeric or tetrameric forms in mucosal secretion
  - monomeric molecule in serum
  - critical role in local immunity

Specific immune response

Antigen → IgM

↓ ← class switching

mediated by T cells & cytokines

IgY or IgA

Strong immune memory

Table 2.1 Properties of chicken immunoglobulin isotypes.

Isotype	Heavy chain (kDa)	Number of H chain Ig domains	Homology to mammalian (%)	Serum concentration (mg/mL)	Sources
IgM	70	5	About 30	1–2	Serum
—	—	—	78 for TM*	—	Cell surface
IgY	67	4	30–35	5–10	Serum egg yolk
IgA	65	4	32–41	~3	Serum
—	—	—	—	—	Bile
—	—	—	—	—	Mucosa (tears, saliva)

Humoral immune response



1st vaccination +
2nd Boost vaccination

= Higher antibody titer

我國現行兒童預防接種時程

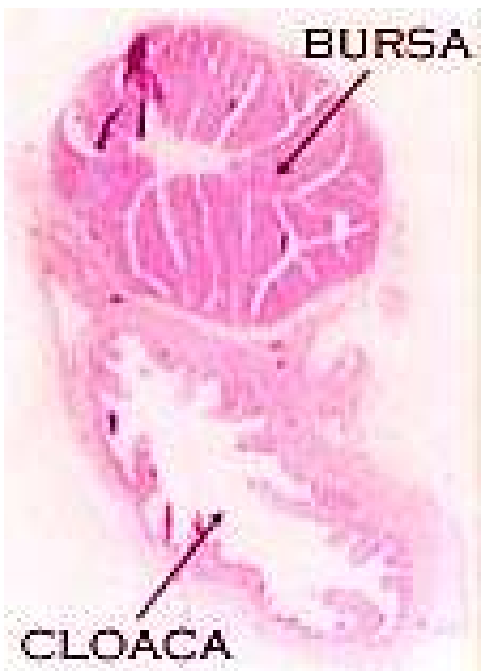
接種年齡 / 疫苗	24hr 內儘速	1 month	2 months	4 months	5 months	6 months	12 months	15 months	18 months	21 months	24 months
B 型肝炎疫苗 (Hepatitis B vaccine)	第一劑	第二劑				第三劑					
卡介苗 (BCG vaccine)[1]					一劑						
白喉破傷風非細胞性百日咳、b 型嗜血桿菌及不活化小兒麻痺五合一疫苗 (DTaP-Hib-IPV)			第一劑	第二劑		第三劑			第四劑		
13 價結合型肺炎鏈球菌疫苗 (PCV13)			第一劑	第二劑			第三劑				

B 細胞的發育 – Chicken

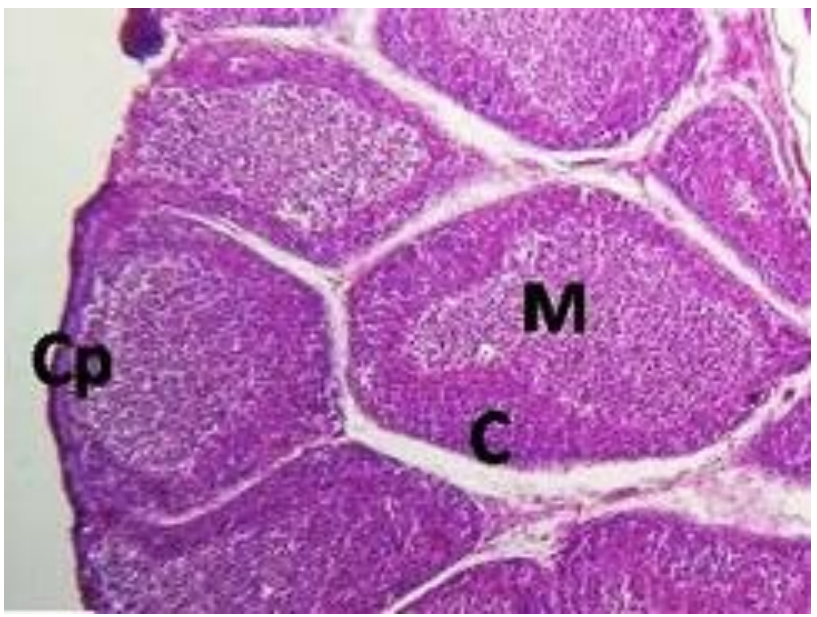
Incubation period for chicks: 21 days (**3 weeks**)

Development of **B cells** in Bursa:

- Follicle formation (cortex, medulla)
- **Gene conversion in B cells → humoral immunity**
- Migration of B cells to peripheral areas
- Embryo development = possible → *in ovo* vaccination



Intact Bursa Fabricius



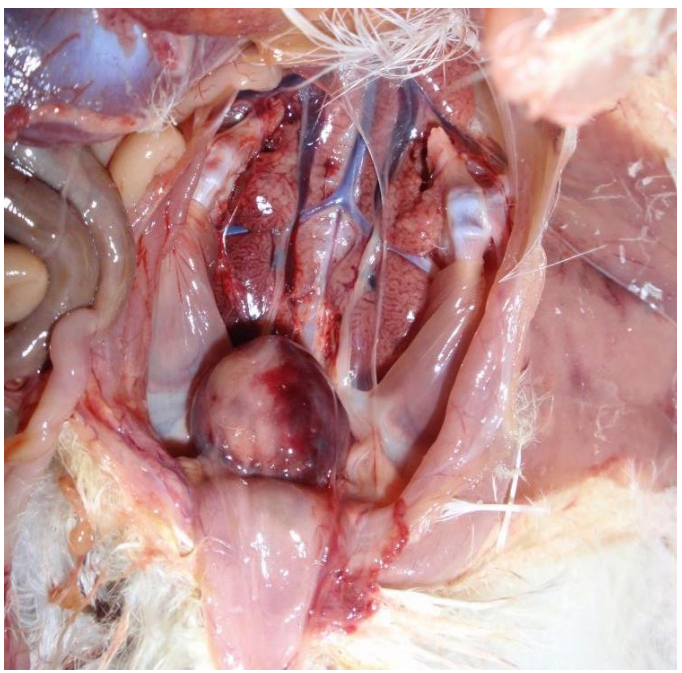
Cp: Capsule; C: Cortex; M: Medulla.

Infectious bursal disease (IBD)

Also named:

Gumboro Disease (甘保羅病)

- first discovered in Gumboro in 1962
- Disease affects Bursa





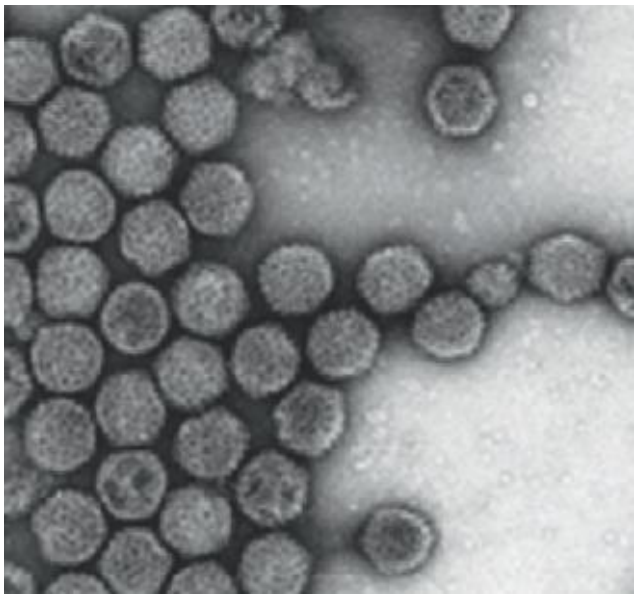
IBD

- IBDV infects chickens, turkeys, ducks, guinea fowl and ostriches but causes clinical disease solely in young chickens.

- 造成嚴重的經濟損失:
  - 3~6週齡雞 急性死亡
  - 2~3週齡小雞 無症狀 但B cell被破壞→免疫抑制

- In Taiwan:
  - 1980 抗體血清學調查: 台灣雞場普遍受IBD汙染
  - 1992 vvIBD台灣大流行

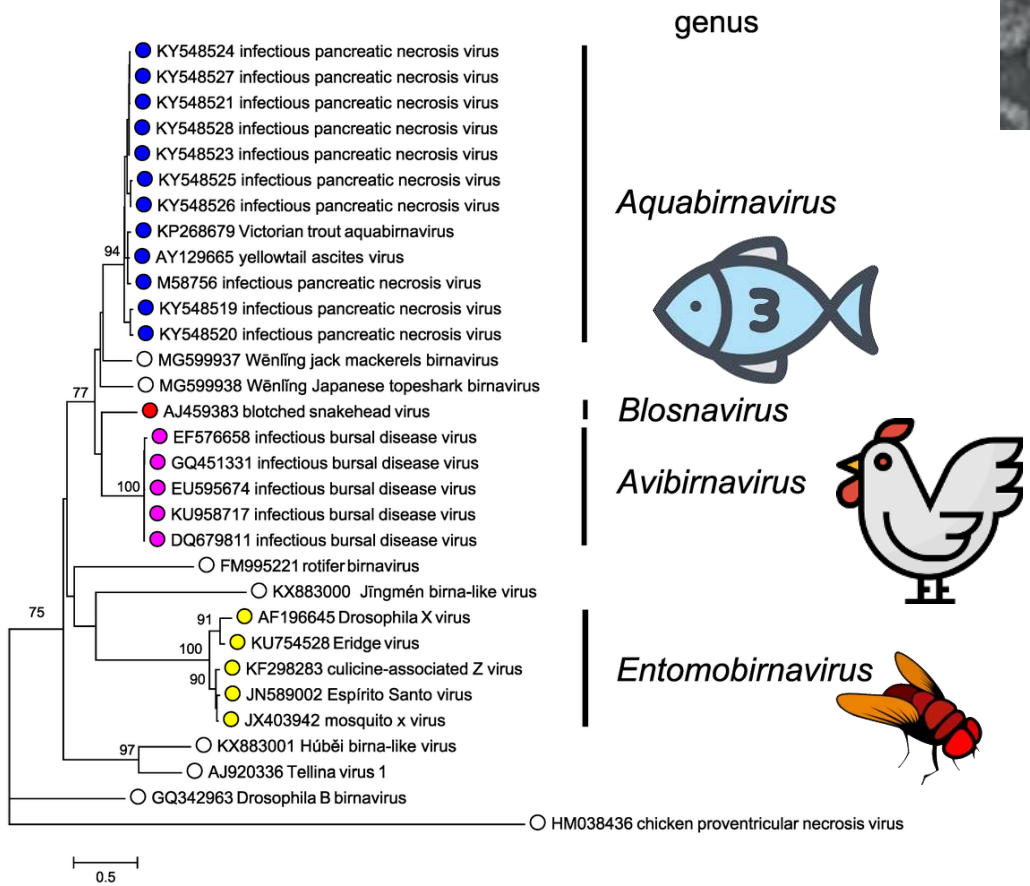
Infectious Bursal Disease Virus (IBDV)



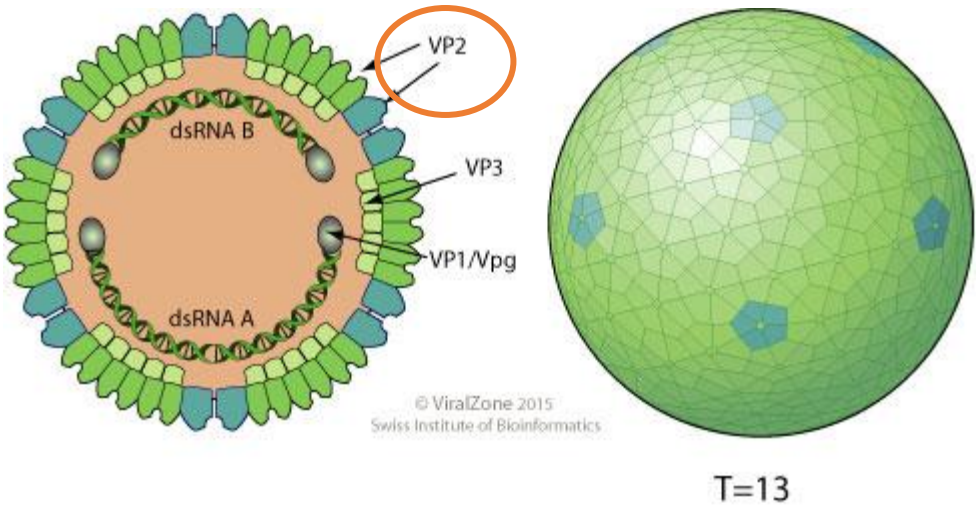
Taxonomy:

Family: *Birnaviridae*

Genus: Avibirnavirus

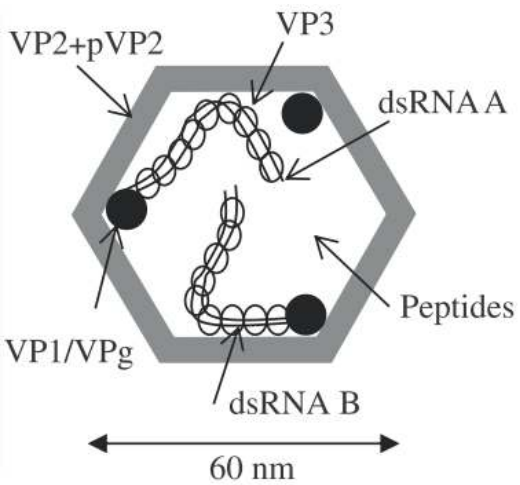


IBDV



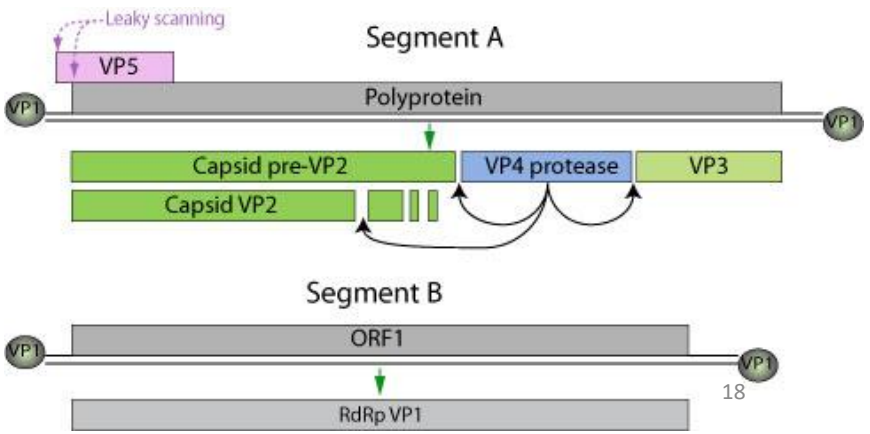
Virus properties:

- non enveloped virion
- Double stranded RNA genome



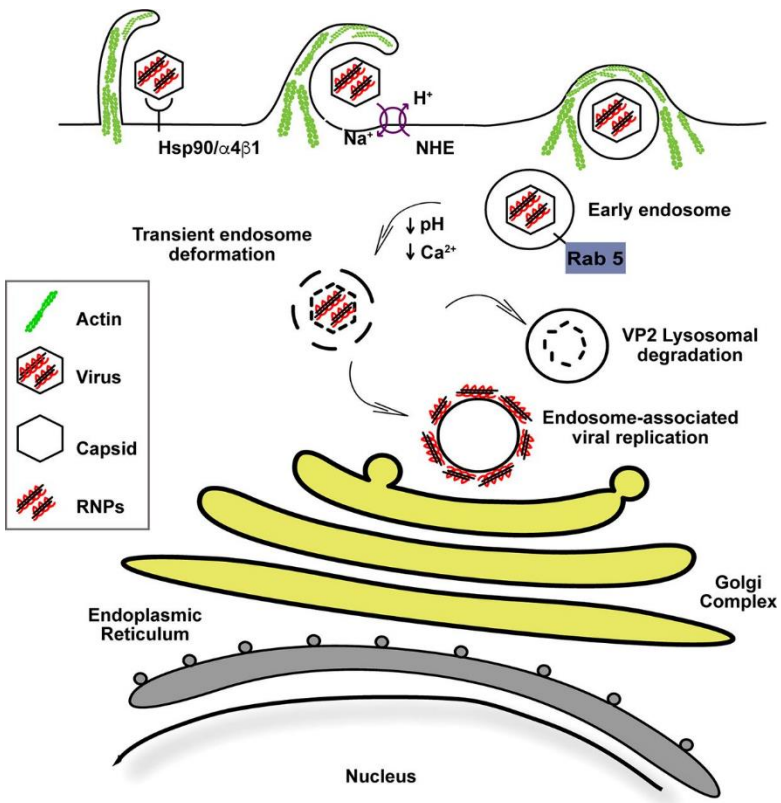
- 60nm diameter
- icosahedral symmetry (20面)
- Two genome segments

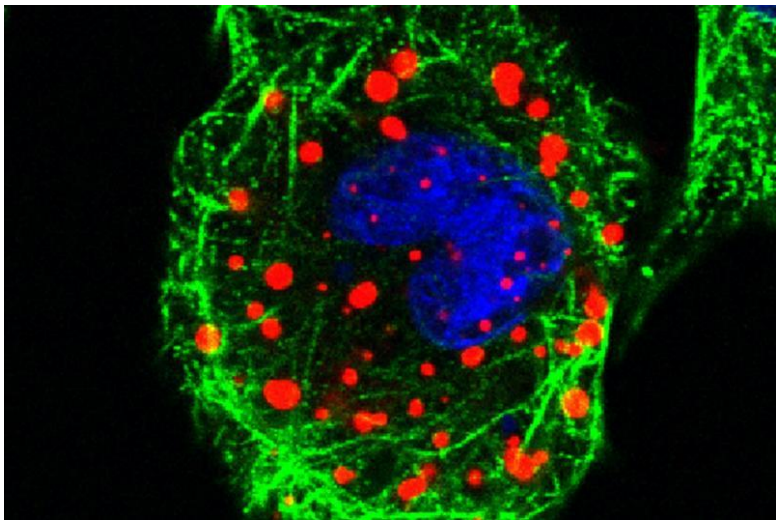
- 對乾燥、日光、部分消毒劑有抵抗性 →消毒水添加碘
- pH 2-11
- 一旦存在於雞舍就不易排除
- 可存在雞舍122天仍有感染力
- 可存在水、糞便、墊料中52天



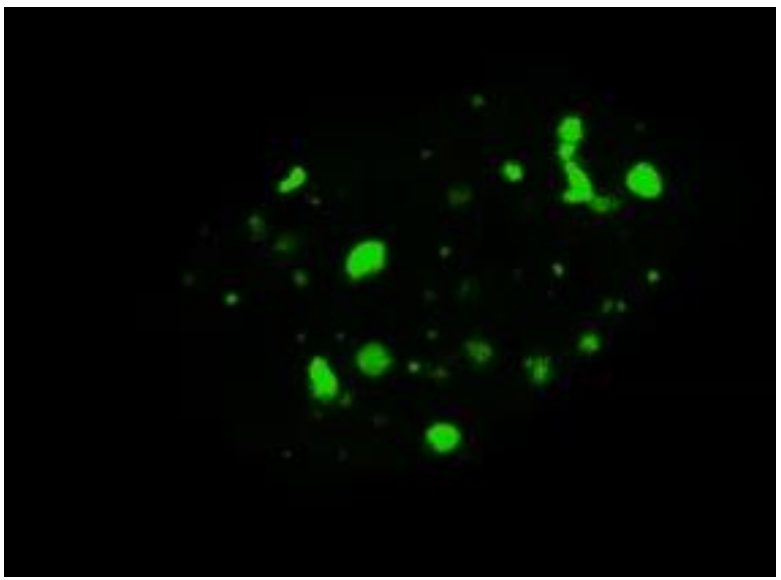
Virus entry/replication

目前已知 – 利用 Endocytosis 進入細胞
在”Cytoplasmic” 之間進行複製





綠-細胞支架tubulin
紅-病毒顆粒



病毒在”Endosome”內複製

IBDV

Serotypes specific，2 serotypes

- Serotype 1 and 2(apathogenic)
- Virulent strains (serotype 1) ，只有雞為自然宿主
- Serotype 2：無致病性

- 兩種血清型間無交叉保護作用

  There is no cross protection between the 2 IBDV serotypes.

IBDV

Virulence dependent

- Very virulent (vvIBDV) “Mortality >20%” (台灣主要致病)
- Classical virulent IBDV “Mortality <20%”

- Moderately attenuated 中間偏強
  - **"Intermediate plus", "hot"**
- Medium attenuated 中間
  - "Intermediate"
- Heavily attenuated 弱
  - "Mild"

Vaccine strains
Attenuated (atIBDV)

- **Variant IBDV strains** antigenic variant (avIBDV)

不表現classical epitopes (抗原性改變)→ 不怕移行抗體
→ 早期感染 → Bursal 萎縮 →immunosuppression.
→ (Mortality < 5%)



https://poultrycontent.ceva.com/

Transmissions

- 經糞便傳播
- 眼、口感染

- 帶病毒糞便汙染雞舍、食物、飲水→經口鼻進入消化道、呼吸道
- 接觸傳播
- 機械性傳播（人車箱籠）

- 將病雞移走後4個月內雞舍殘留的病毒仍可感染新的雞隻



Non-enveloped Virus → **resistance** to disinfection (70% EtOH …etc)

MDA = Maternal Derived Antibody 移行抗體

Pathogenesis

- Infection is the cause of **immunosuppression** within the first **3 weeks**

- Within 3-6 weeks, the clinical form of the disease occurs:
  - 一般3-20週齡間發病，多發生在3-6週齡的雞
  - Clinical findings 2-3 days after infection 潛伏期2-3天
  - Mortality 5-8 days after infection

- IBDV infects cells of the cloacal **bursa** – primarily immature B-cells.
  - mature B-cells = antibody producing cells
  - If B-cells are destroyed early in life the bird may become **immunosuppressed for life.**

Clinical signs

- 急性型: 3-6週齡，傳播速度快，急性發病2-3日死亡
  - 精神沉鬱、食慾差、羽毛蓬鬆、飼料換肉率差
  - 水樣下痢便→墊料潮濕、脫水、病情嚴重會無法站立
  - 血樣下痢 (華氏囊出血)
  - 啄肛→尾巴變短、沒有尾巴
  - 尾毛濕、尾毛掉、尾部出血、腸被拉出

- 免疫抑制型: 小於3週齡，不常見臨床症狀
  - 華氏囊病變、萎縮→免疫抑制→二次性感染、降低其他疫苗效力
  - 生長差、疫苗失效

Clinical signs



Clinical signs







Mortality rate

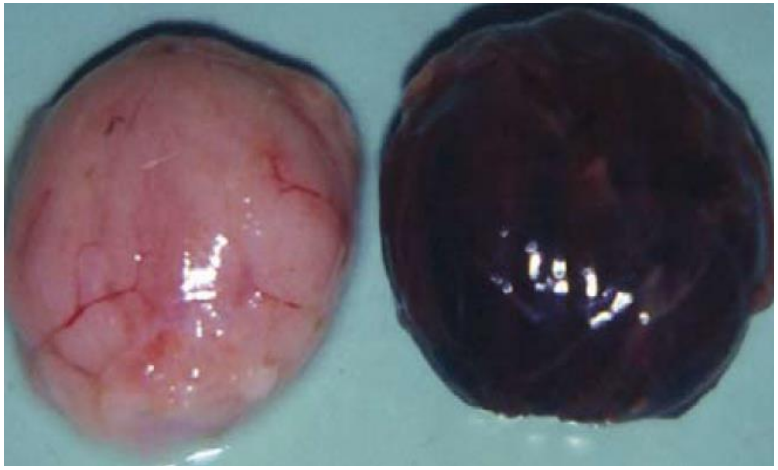


	vvIBDV	Classical IBDV	Variant IBDV
Mortality rate	>20%	<20%	<5%

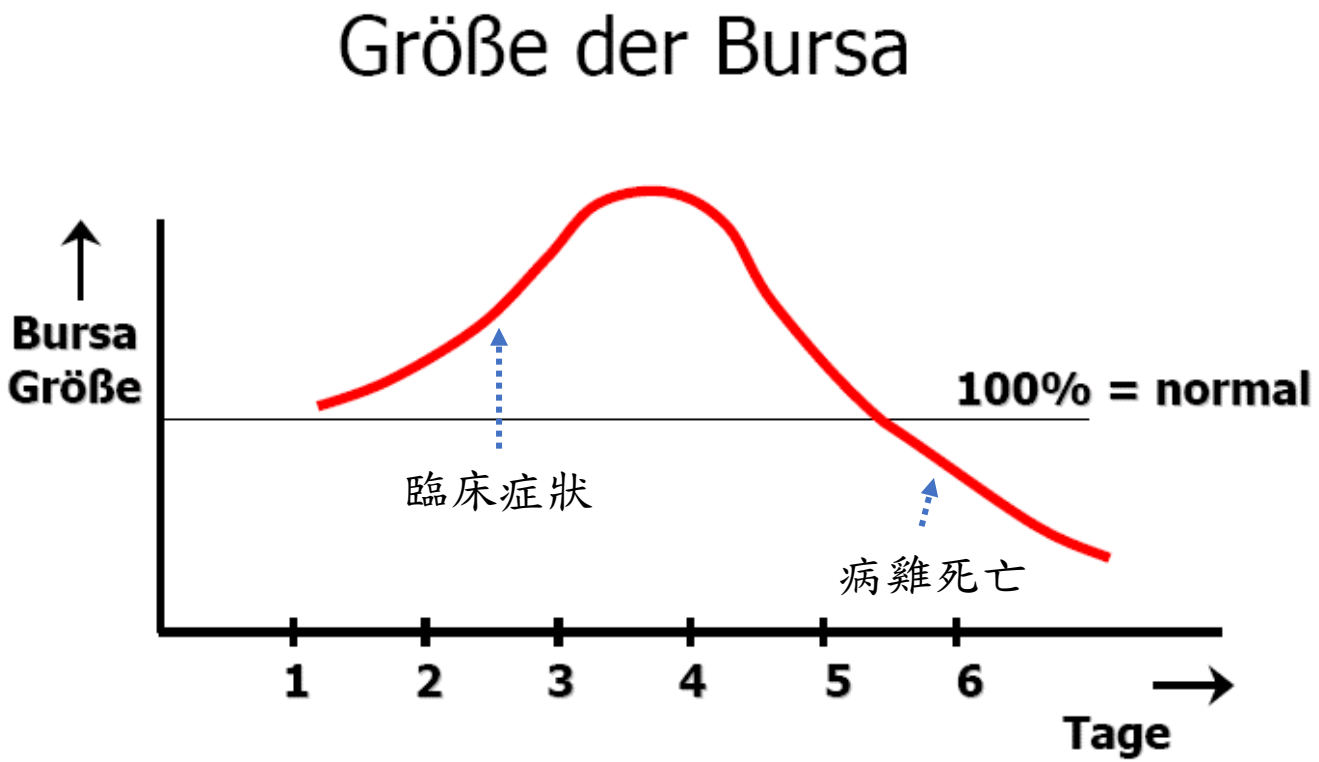
肉眼病變



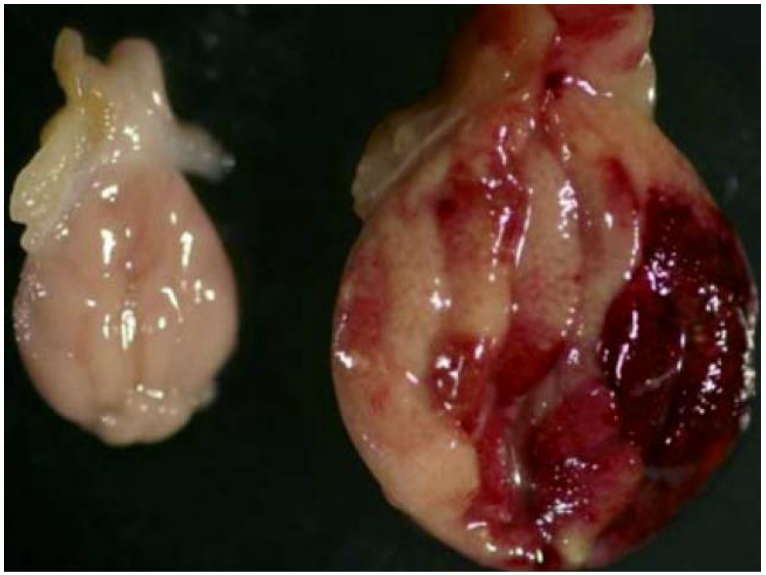


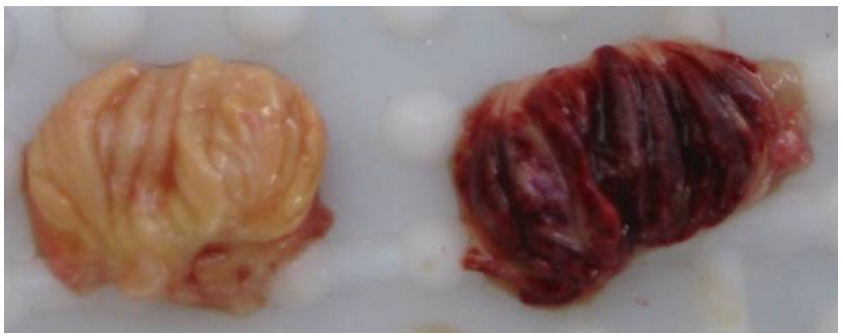


Bursa 大小變化


Größe der Bursa
Bursa
Größe
100% = normal
臨床症狀
病雞死亡
1
2
3
4
5
6
Tage

程度不一的肉眼病變

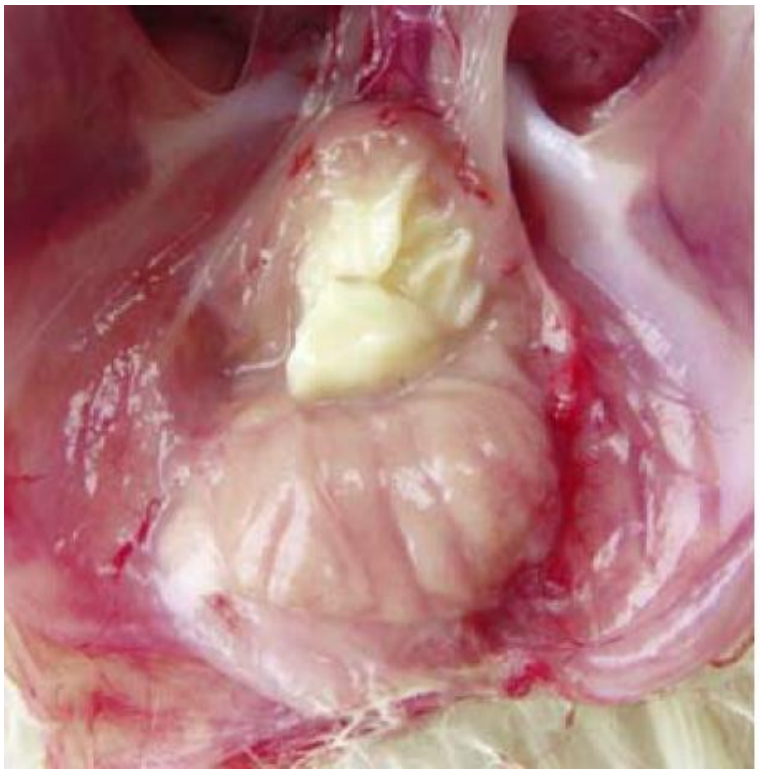


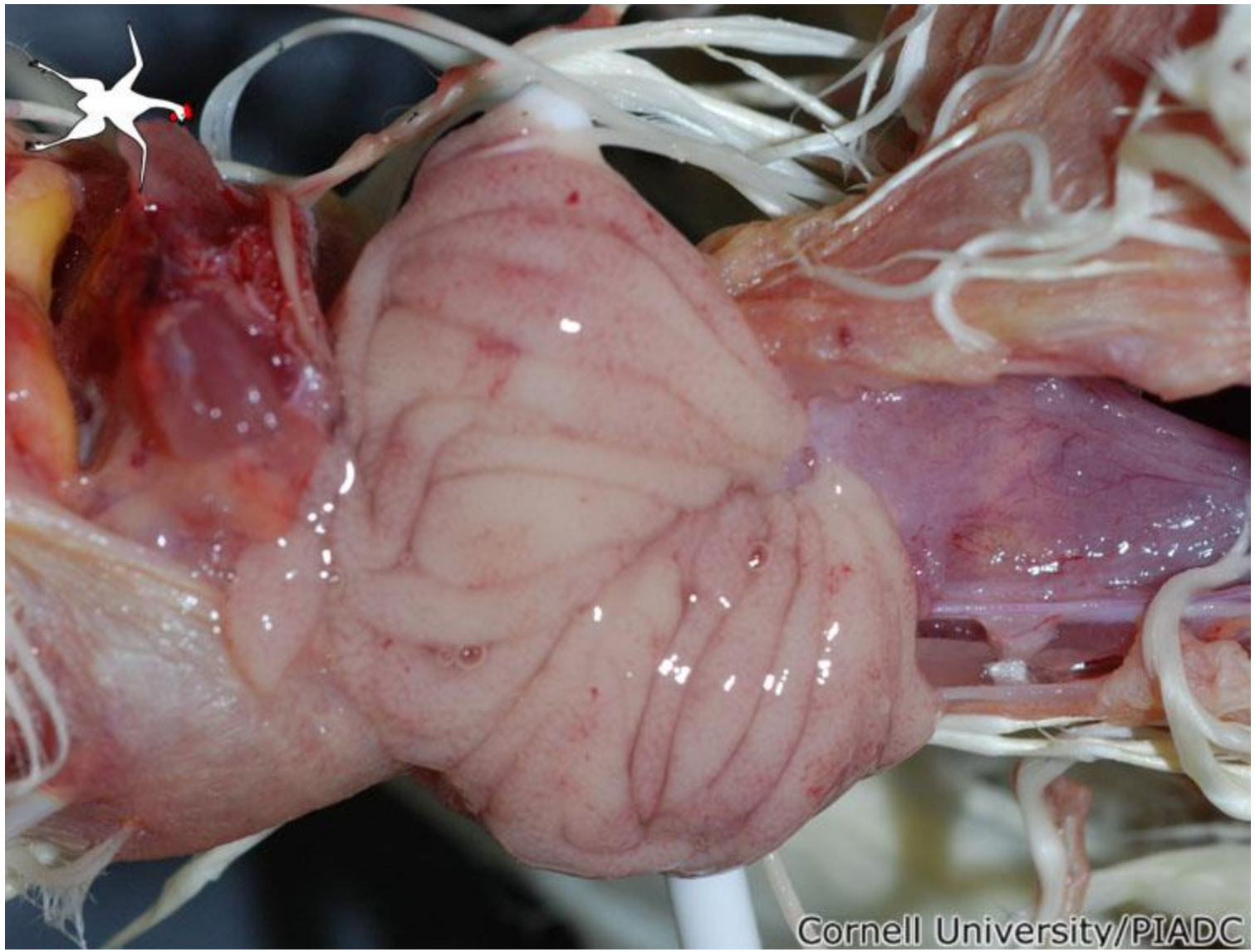


Bursa 水腫　出血

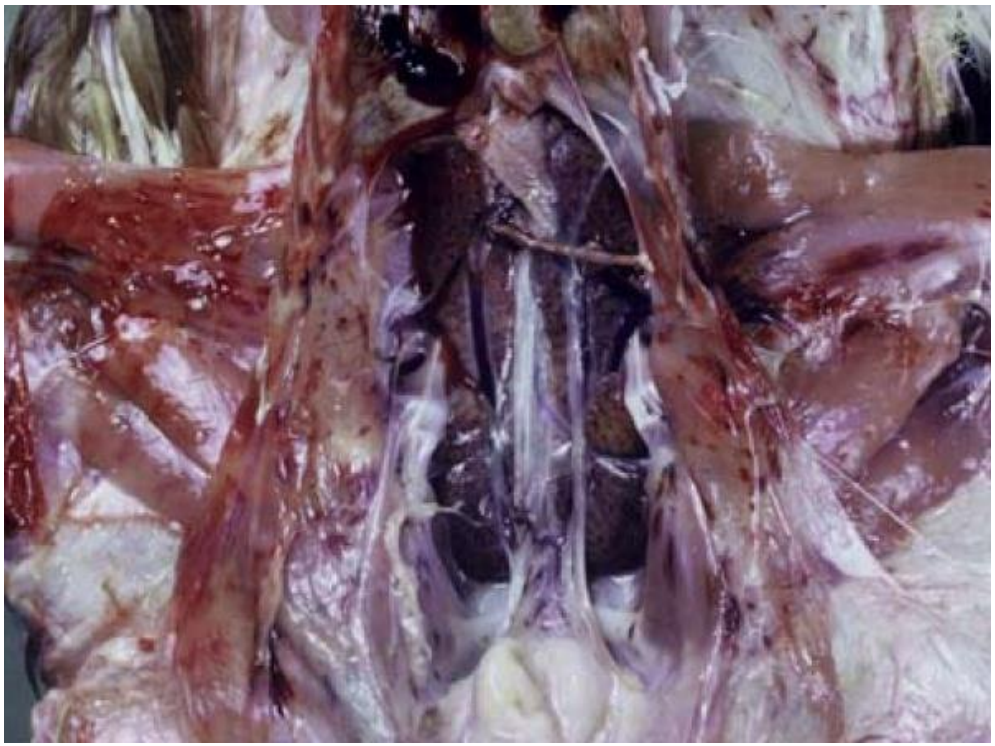


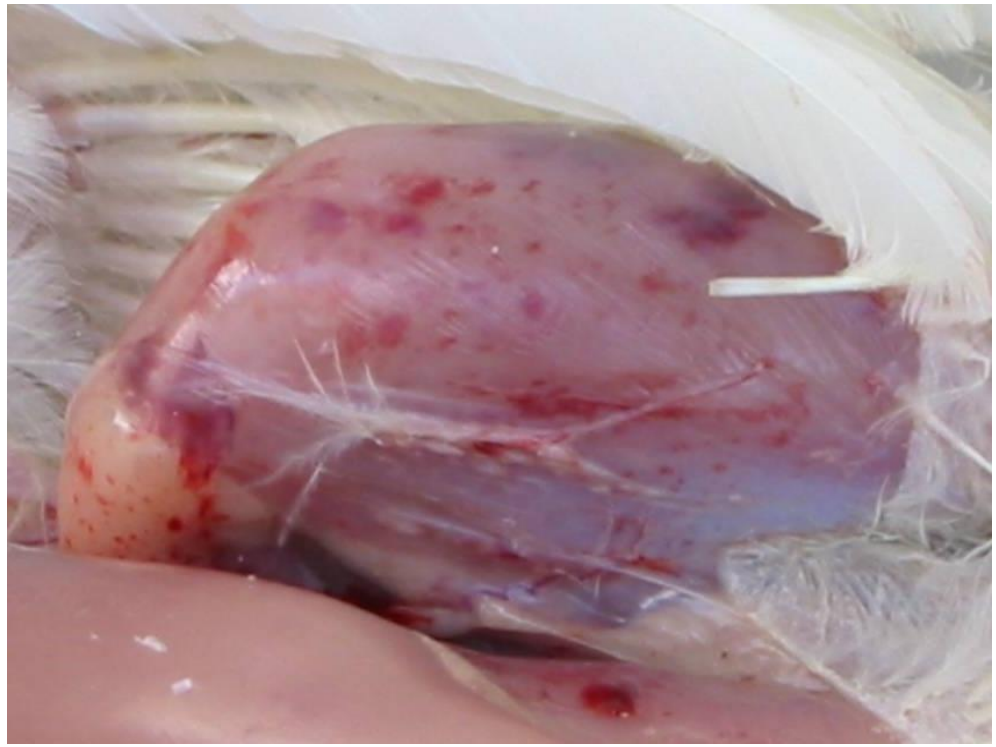
Petechial hemorrhages




Cornell University/PIADC

肉眼病變



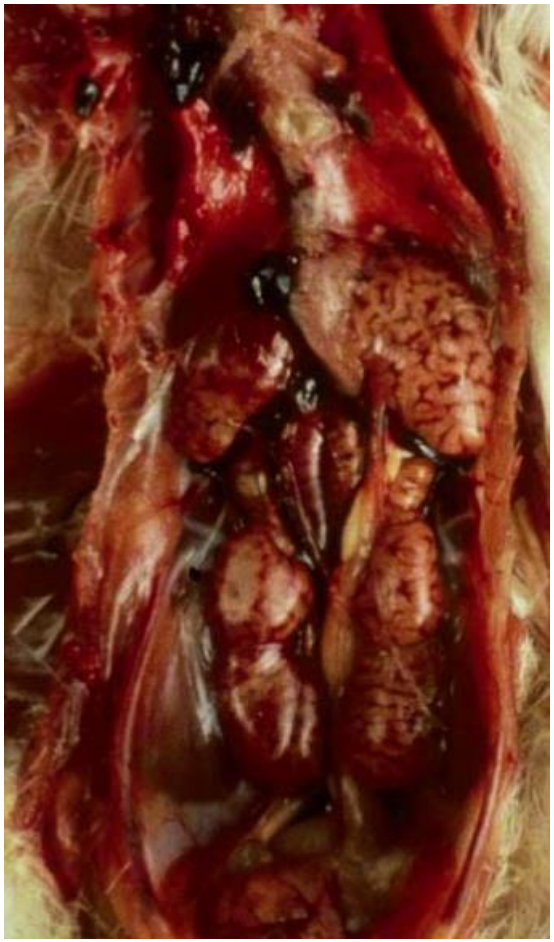


The dead birds are dehydrated, often with **hemorrhages** in the pectoral, thigh and abdominal muscles.

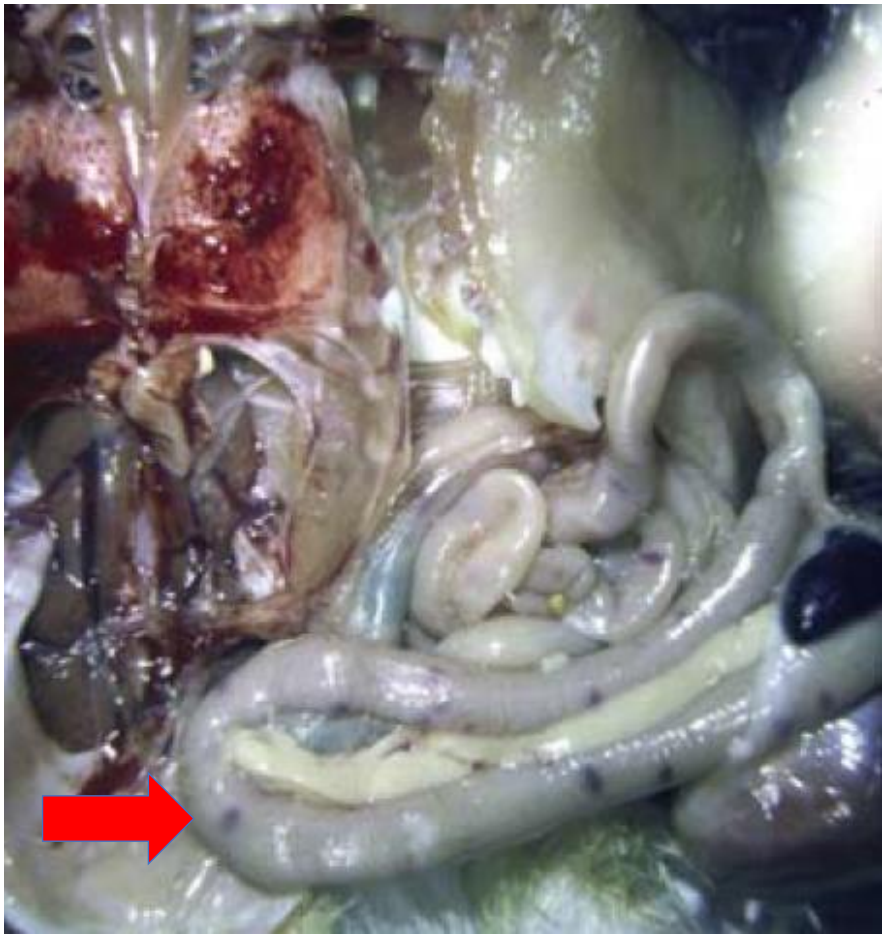
肉眼病變



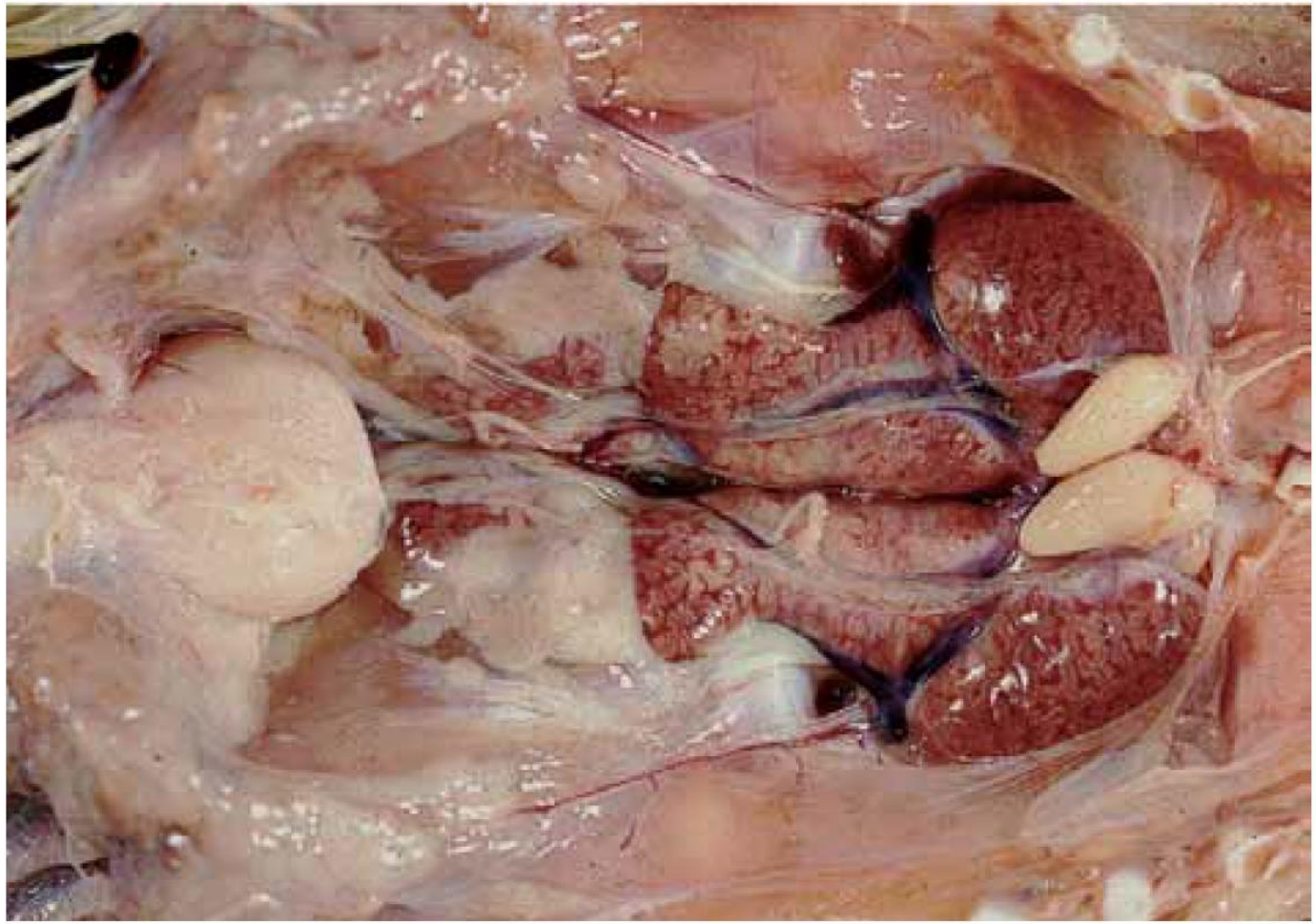
Normal chicken



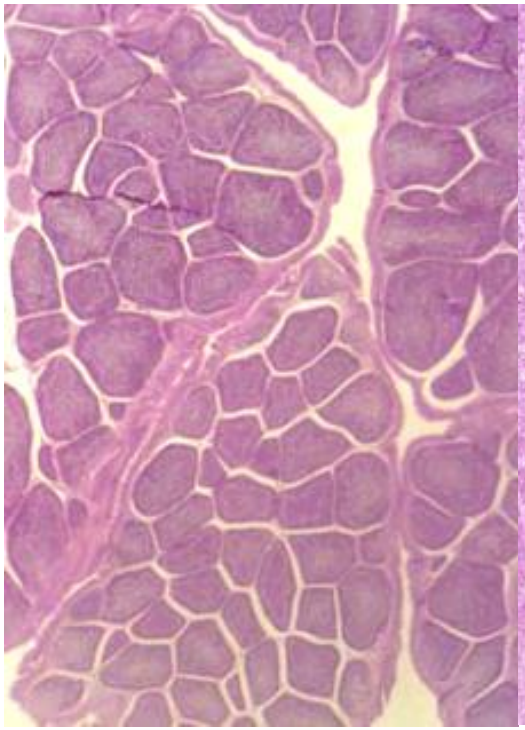
IBDV infected chicken kidneys appear swollen



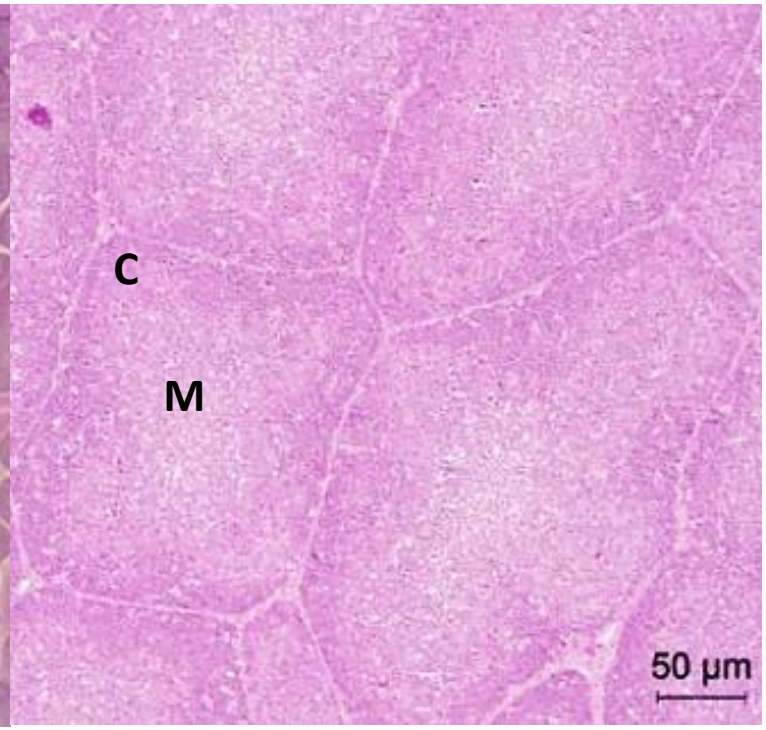
Duodenum hemorrhage



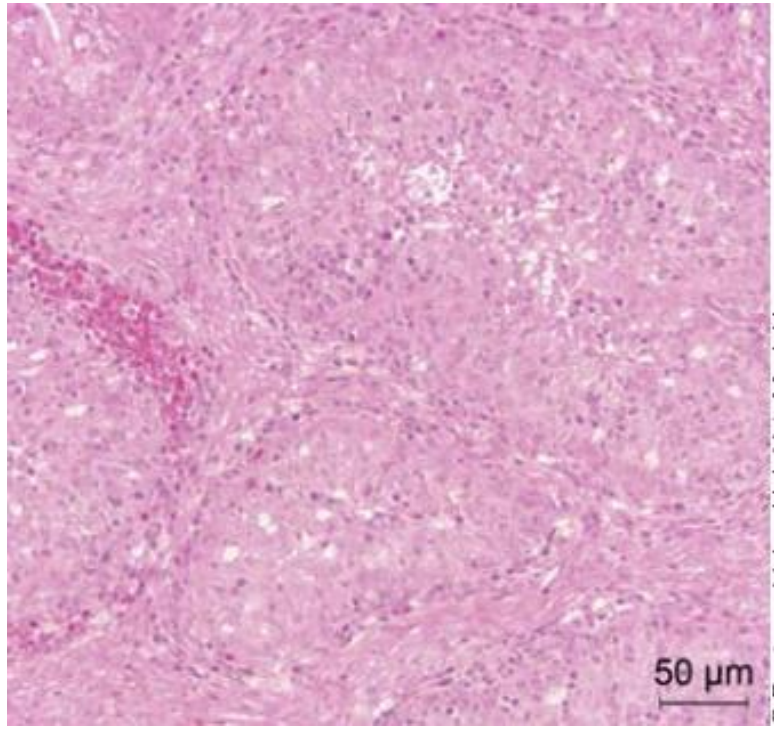
組織病理學



正常Bursa 組織
完整 Follicle

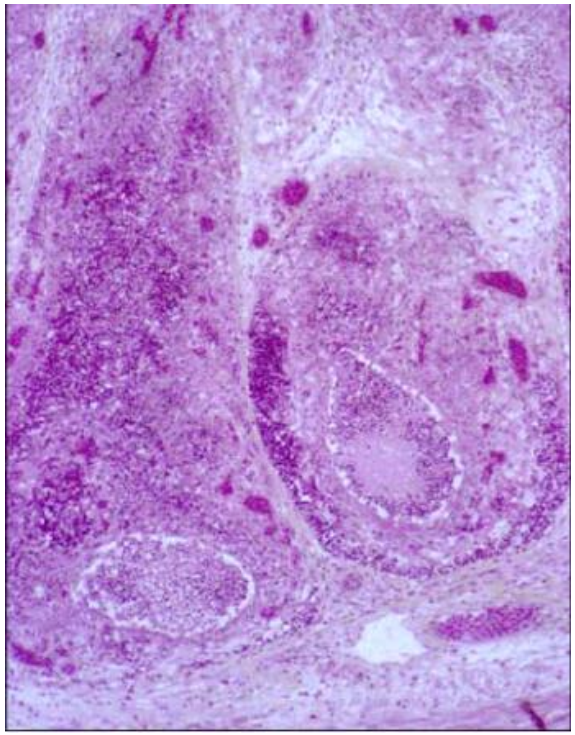


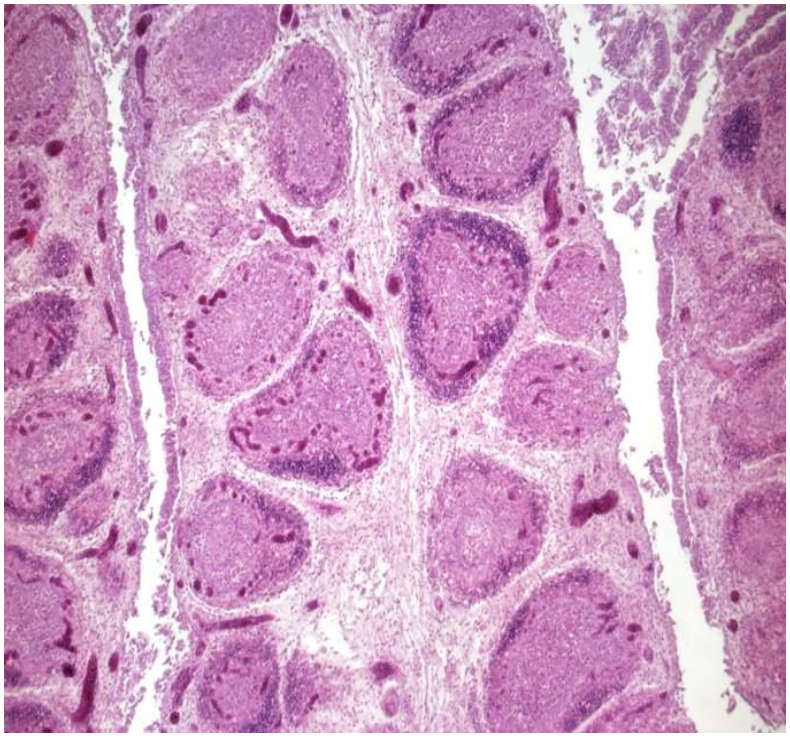
C: Cortex; M: Medulla



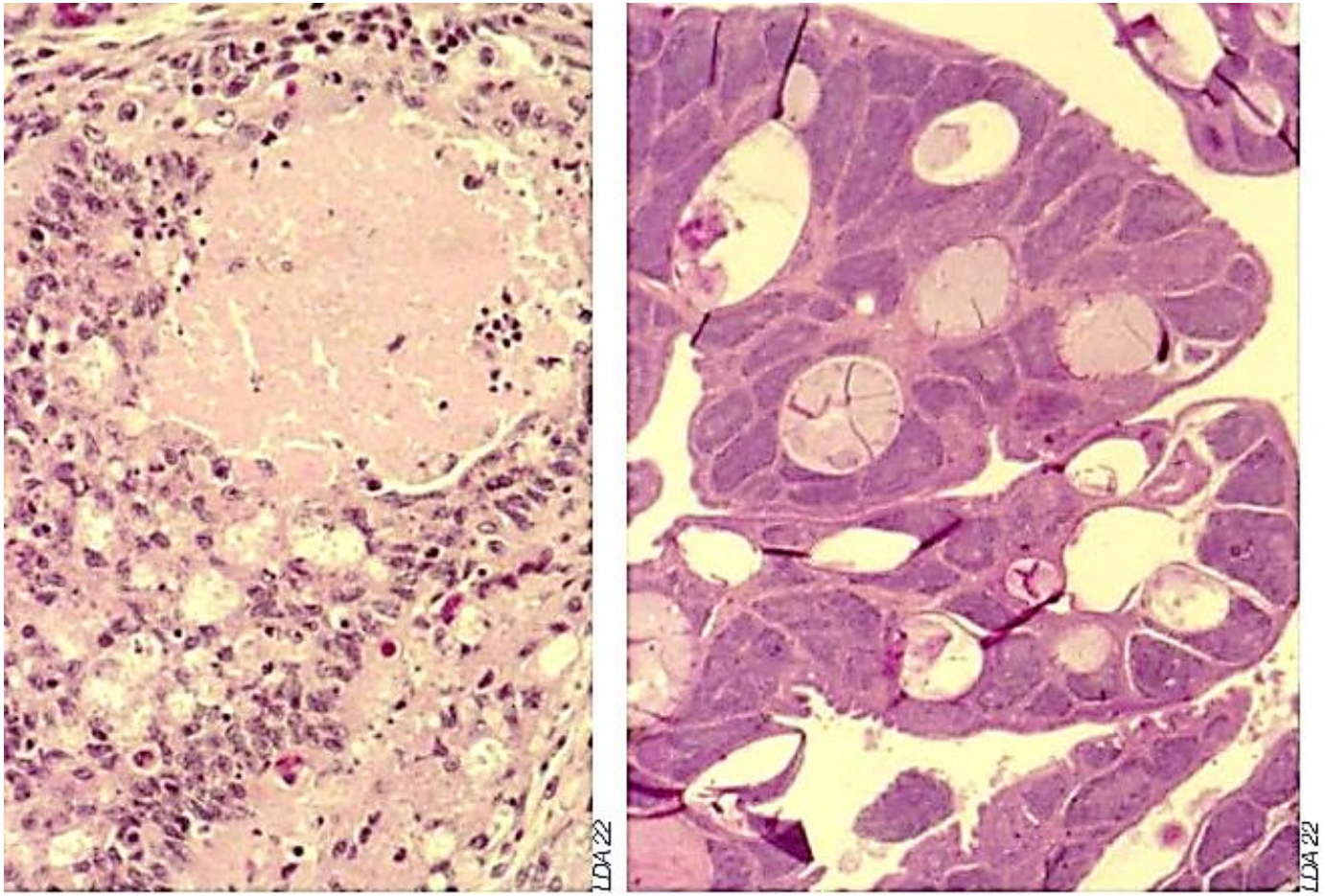
IBDV infected chicken

組織病理學





組織病理學



Massive destruction of lymphoid cells in the bursal follicles.

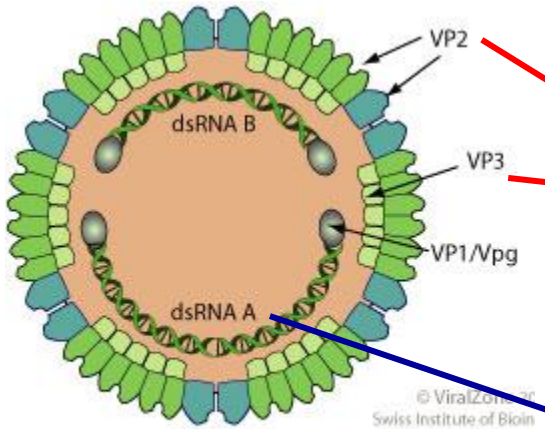
→ **Cysts** formation within the follicles

Diagnosis

Table 1. *Test methods available for IBD diagnosis and their purpose*

Method	Purpose					
	Population freedom from infection	Individual animal freedom from infection prior to movement	Contribute to eradication policies	病例診斷	Prevalence of infection – surveillance	Immune status in individual animals or populations post-vaccination
Pathology and virus detection[1]						
Histopathological examination of bursae	+[a]	–	–	+++	+[a]	+[b]
Virus isolation	+[a]	–[c]	–	+[d]	+[d]	–
Virus characterisation (pathotyping, antigenicity, nucleotide sequencing)	+[e]	–	–	+++	+[e]	+[b]
Virus detection in the bursa by immunoassays (AGID, AC-ELISA, immunostaining)	+[a]	–[c]	–	+++	+	–
Virus detection by RT-PCR	+[a]	–[c]	+[a]	+++	–[f]	+
Detection of immune response						
AGID for antibody detection	++[a]	++	++	–	–	+
ELISA for antibody detection	+++[a]	+++	+++	–	–	+++
Virus neutralisation	+[g]	++[g]	–	–	–	++[g]

Laboratory diagnosis

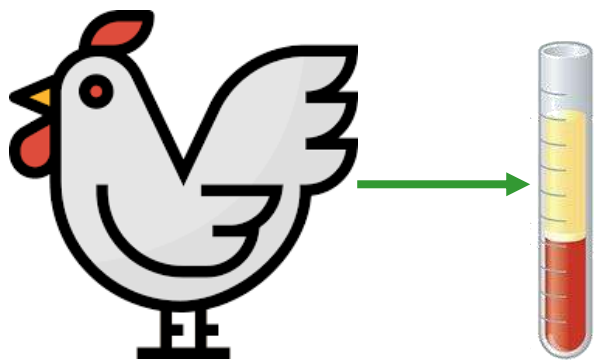


CAPSID ANTIGENS

Characterisation by Monoclonal Antibodies (mAb)

Viral RNA:

RT-PCR or RFLP



Antibody:

ELISA, AGID
Neutralization assay

病毒分離非必要
RT-PCR 及 RFLP較常使用

Virus isolation

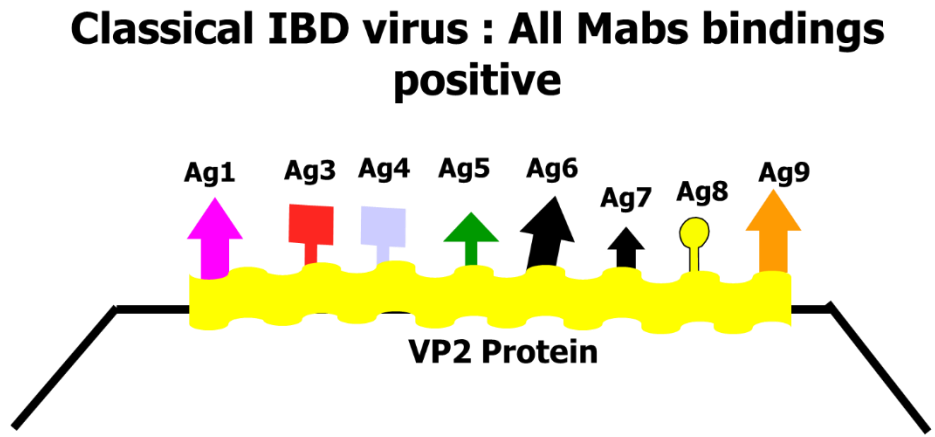
- Bursa: acute phase of the disease
- CAM: 9-11 days SPF eggs
  - 3 days post-inoculation (dpi): death
  - hemorrhages and edema
- Primary bursa cells
  - not recommended for vvIBDV
- SPF chickens
  - the best method for virus isolation
  - virus is then isolated from the bursal tissue 3 dpi

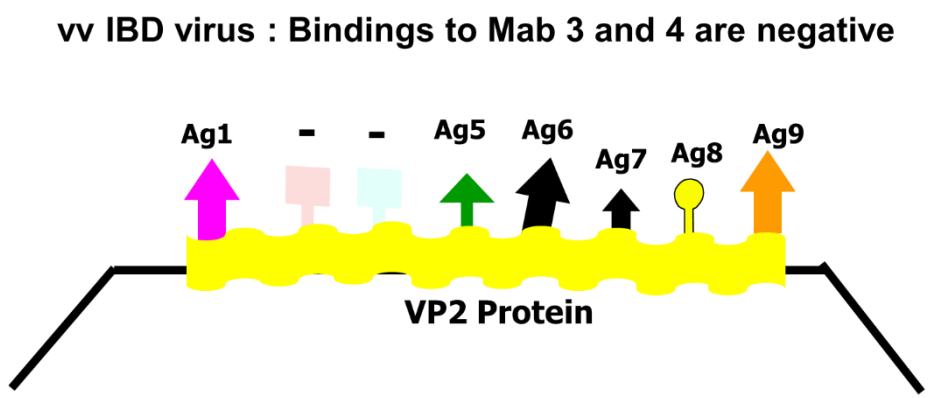
Antigen-capture enzyme-linked immunosorbent assays (AC-ELISA)

病毒分型方法之一

利用病毒Capsid上之**表面抗原**

配合 8株單株抗體製成的
Sandwich ELISA 分“病毒類型”



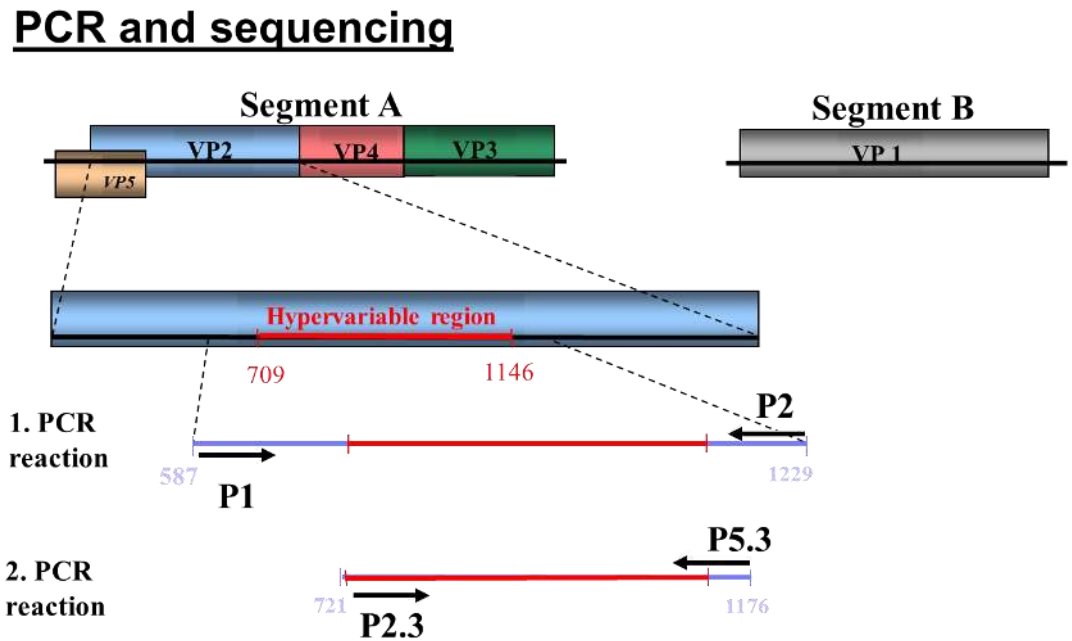


舉例: 有些vvIBDV缺少 Antigen 3 及 4

RT-PCR for the viral RNA

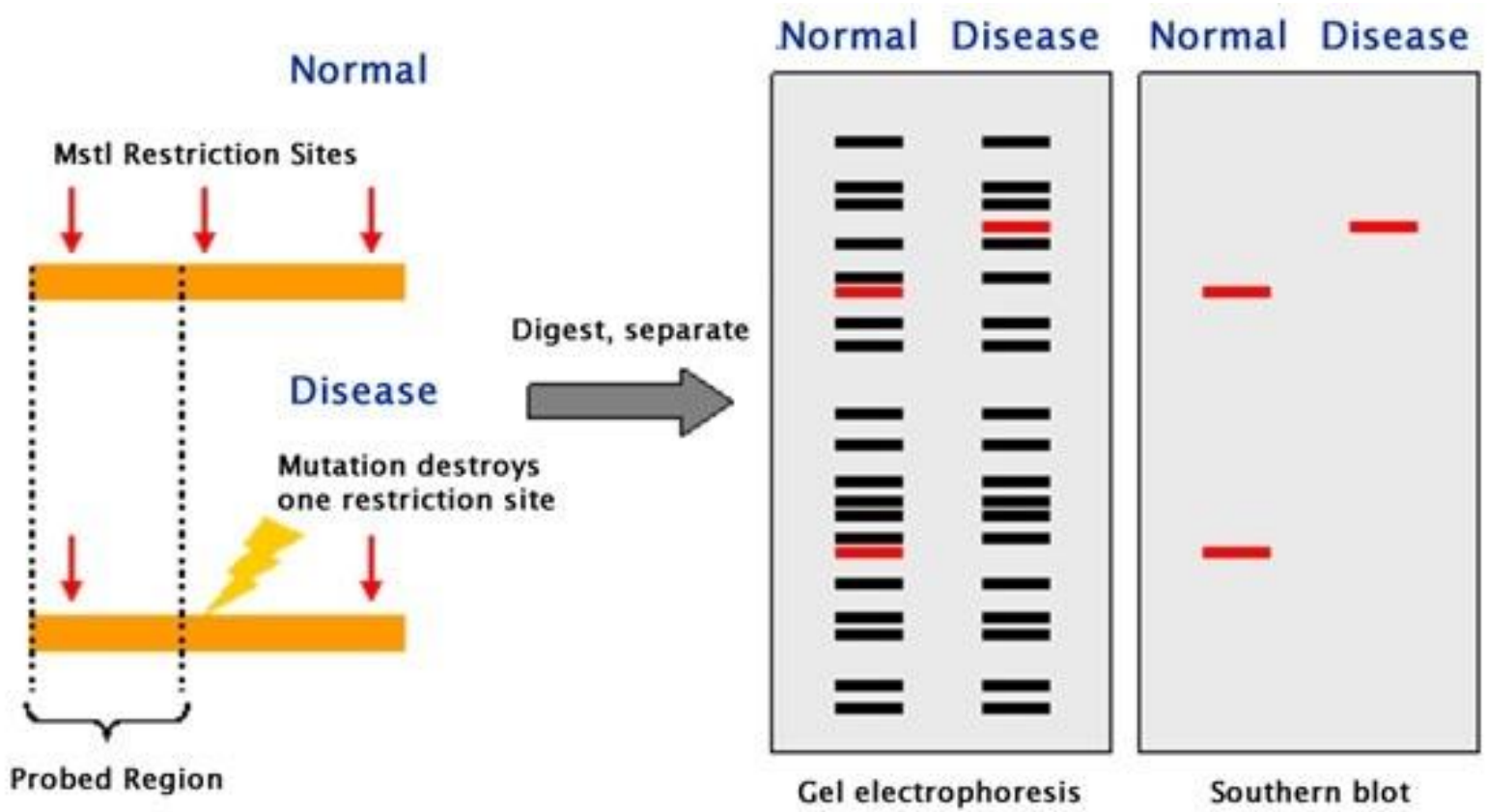
vvIBDV
classical IBDV
Variant IBDV

Attenuated IBDV:
- Intermediate Plus
- Intermediate
- Attenuated



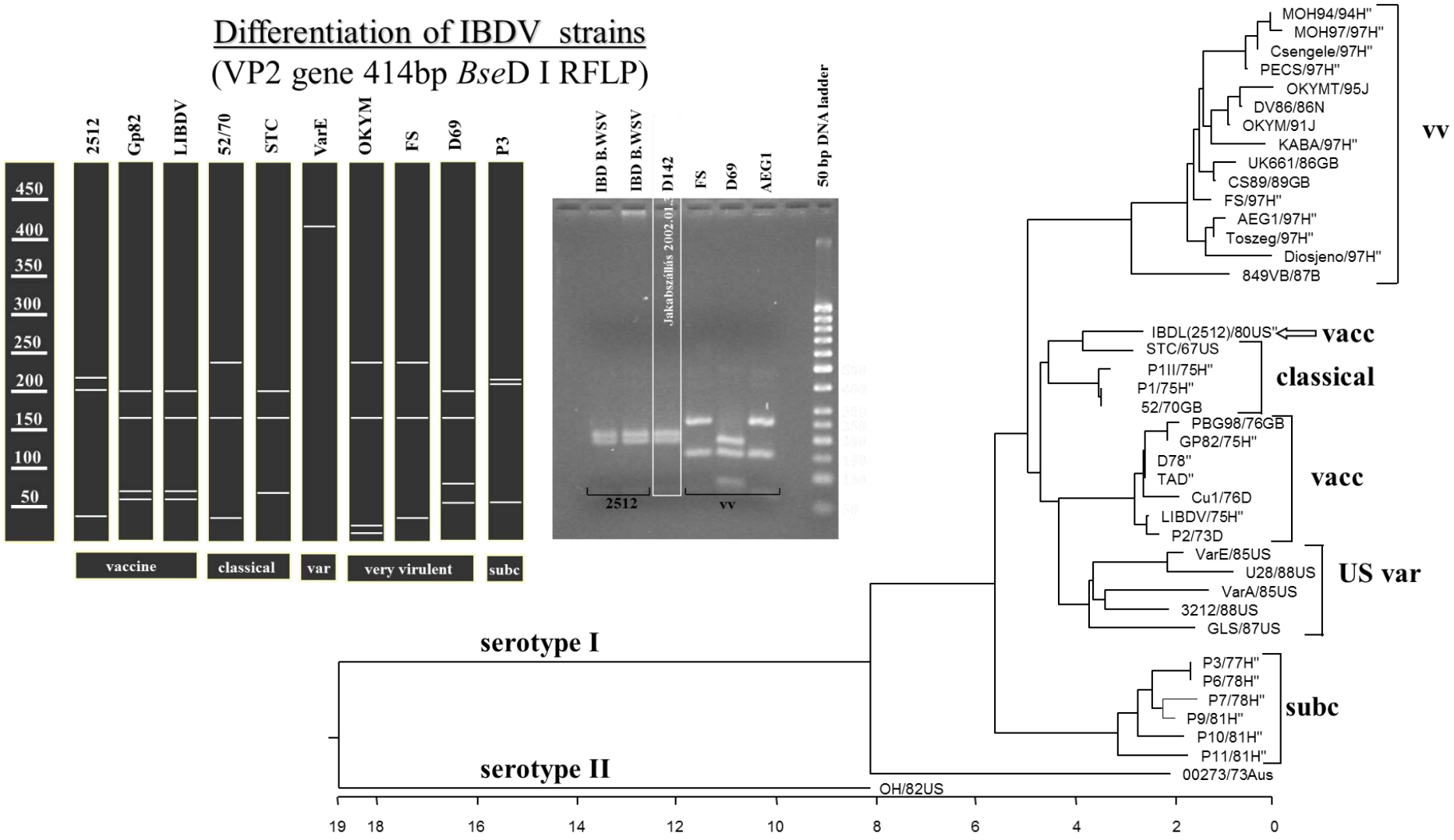
Common RT-PCR diagnosis is not enough

RFLP, restriction fragment length polymorphism
限制性片段長度多樣性



https://www.news-medical.net/life-sciences/Restriction-Fragment-Length-Polymorphism-(RFLP)-Technique.aspx

RFLP, restriction fragment length polymorphism
限制性片段長度多樣性



Agar gel immunodiffusion (AGID)

Technique to identify the antibody-antigen pairing

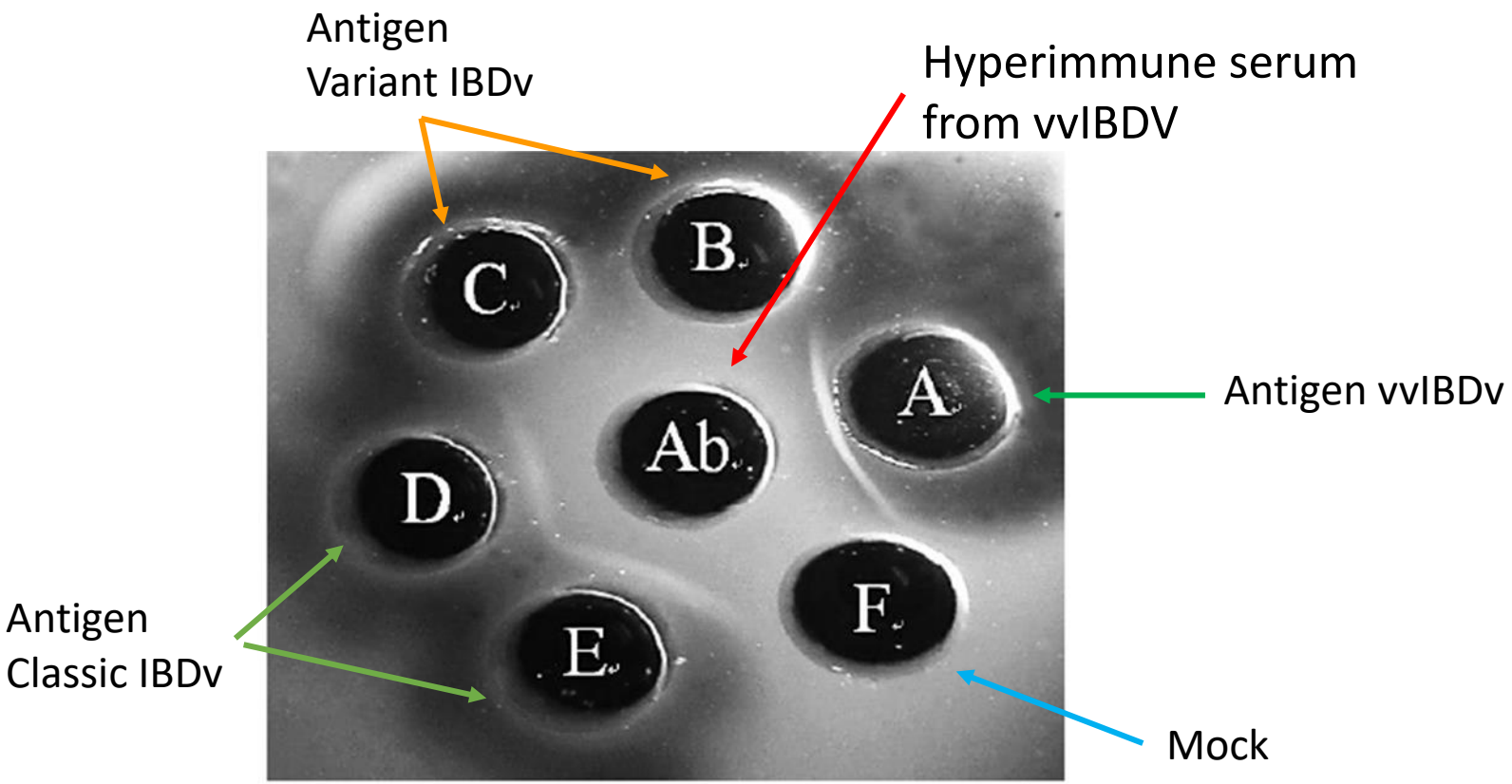


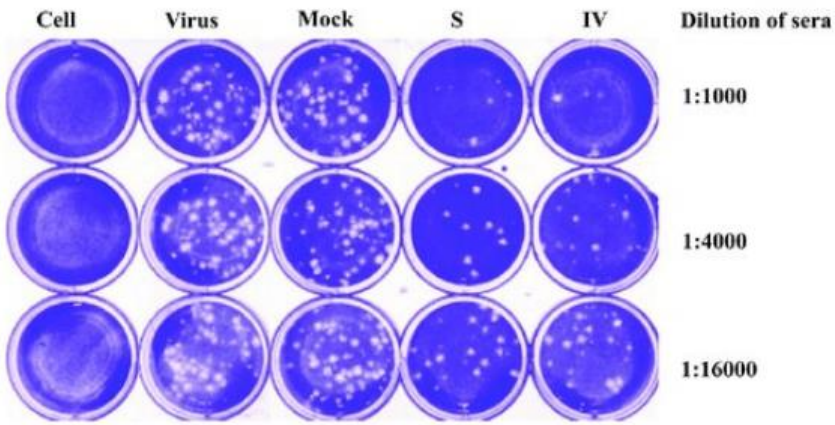
Figure 1. ODIT indicating the occurrence of IBD virus in bursal samples using hyperimmune serum. Ab = Hyperimmune serum against IBDV, A = haemorrhagic bursal suspension, B = atrophied bursal suspension 1, C = atrophied bursal suspension 2, D = oedematous bursal suspension 1, E = oedematous bursal suspension 2, F = mock solution.

(antibody) ELISA vs Virus Neutralization

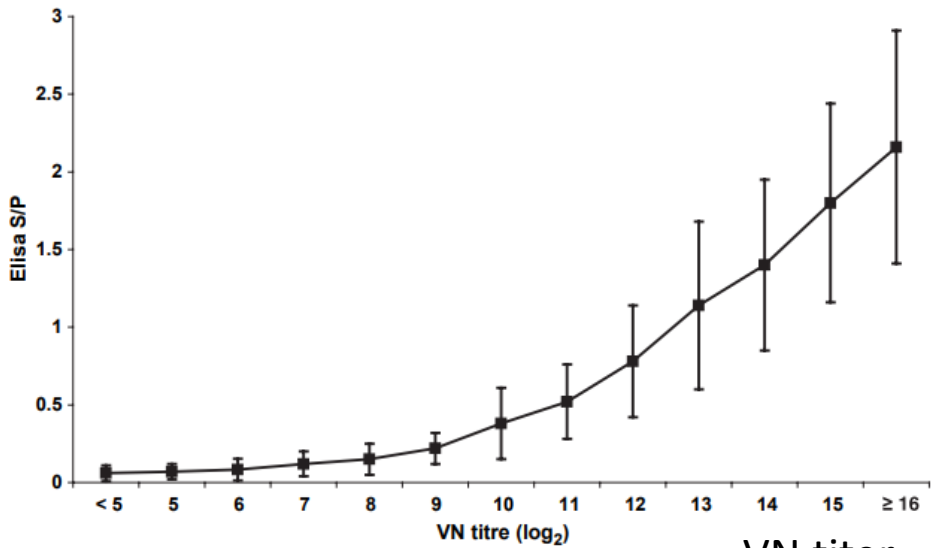


vs

中和試験



ELISA titer



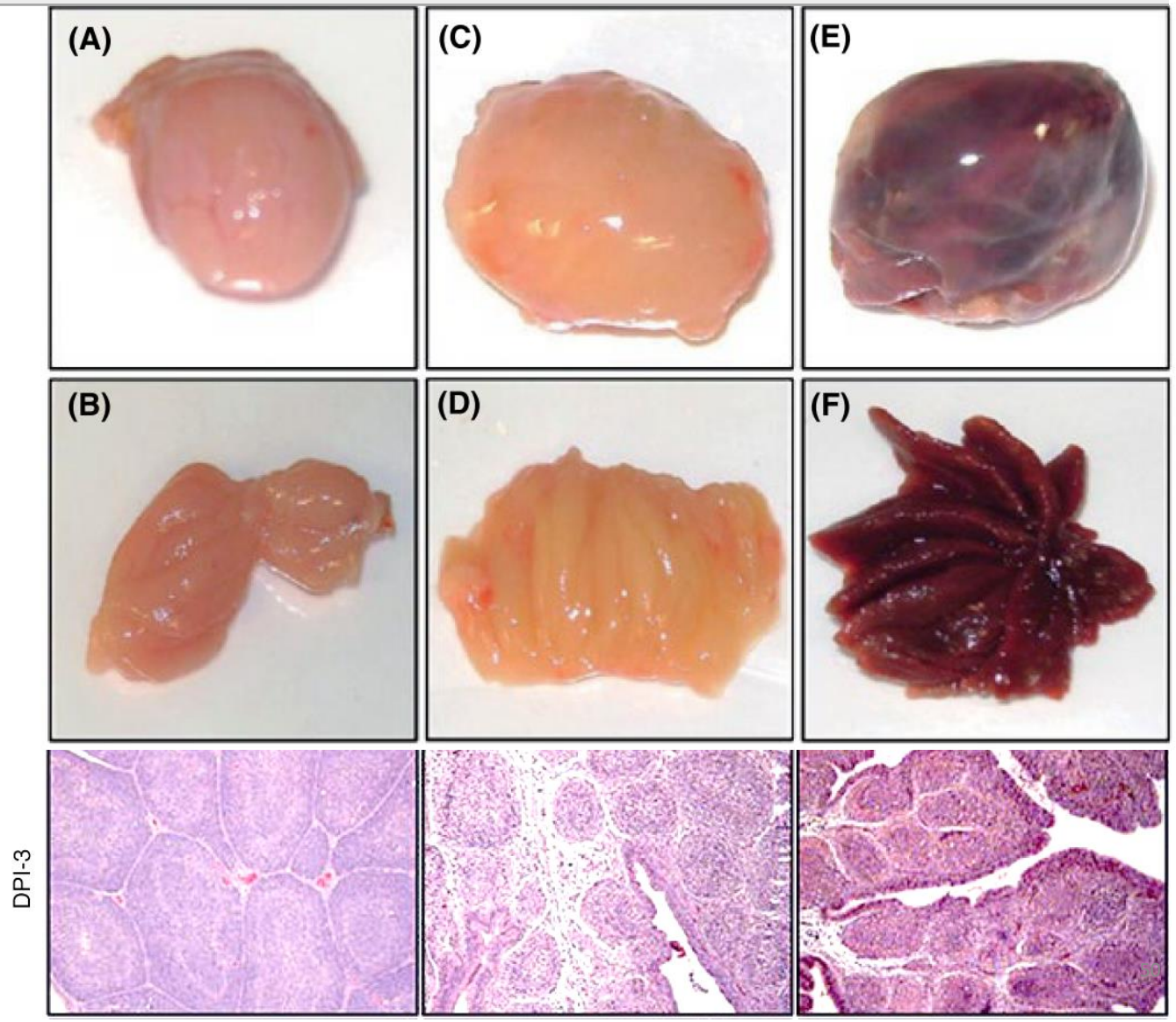
VN titer

Virulence differences

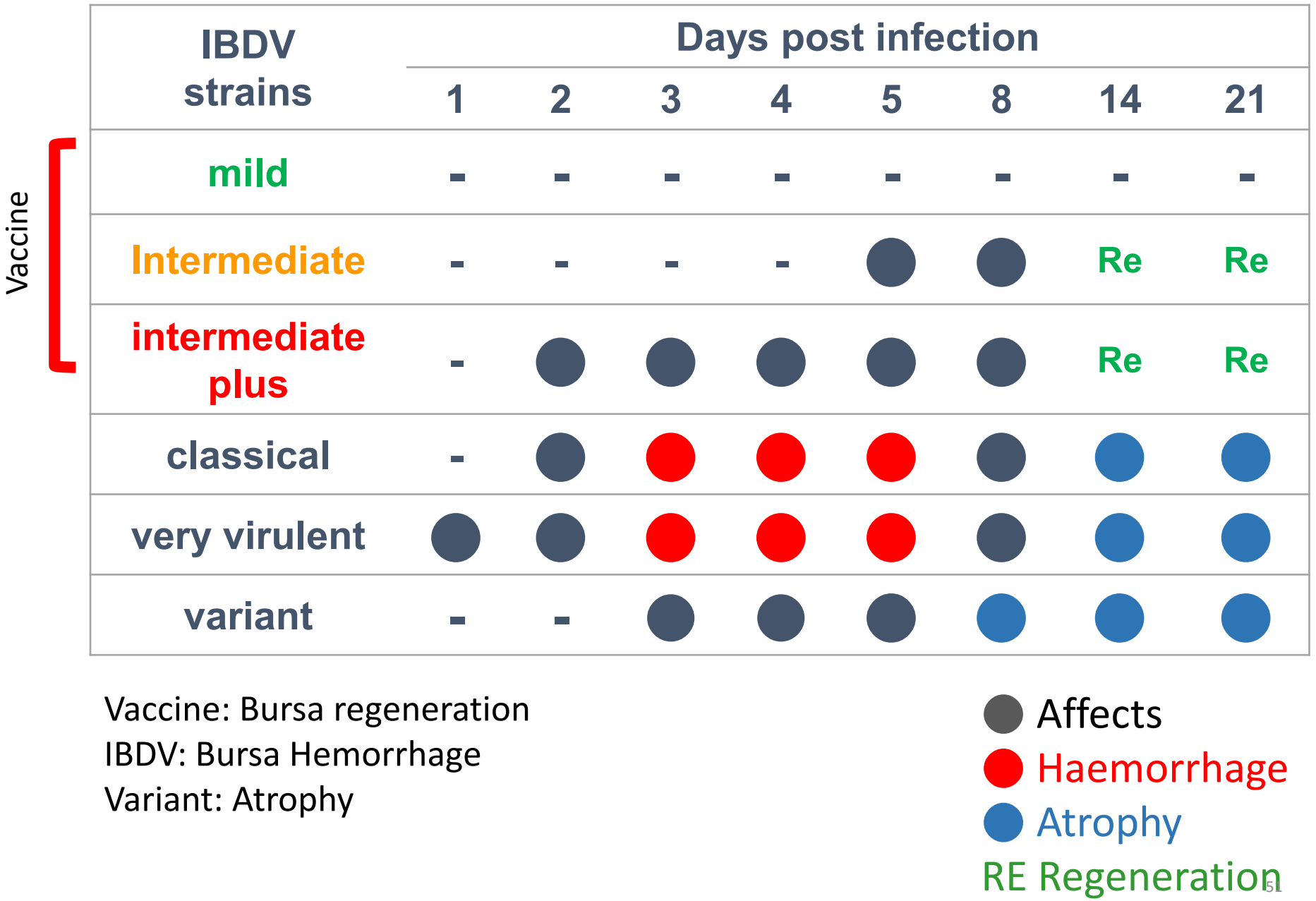
	Non infected	Variant IBDV	Classical IBDV
Mortality	0%	<5%	**20%**

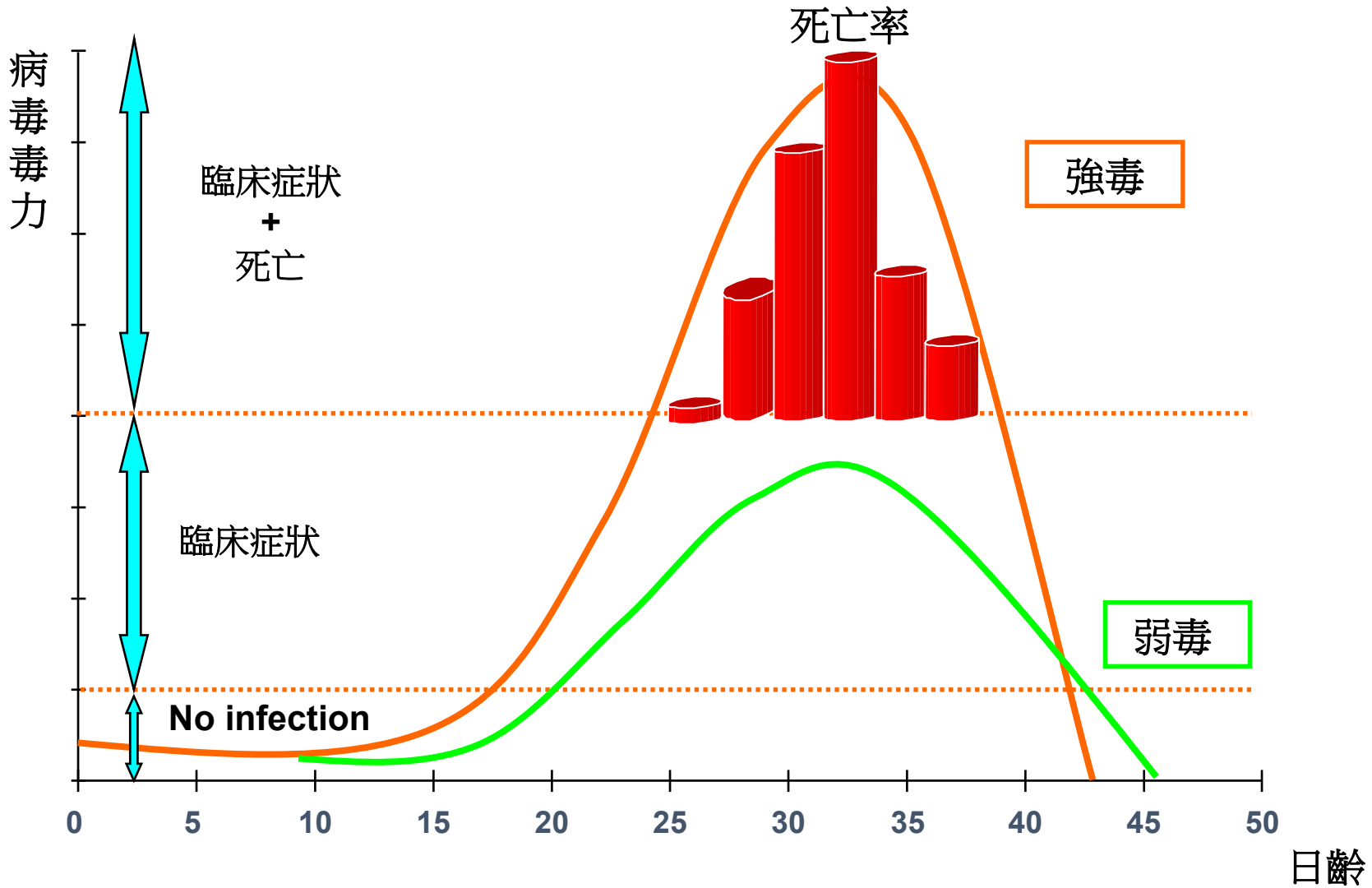
Many IBDV

vvIBDV
classical IBDV
Variant IBDV



Virulence differences




死亡率
病毒毒力
臨床症狀
+
死亡
強毒
臨床症狀
弱毒
No infection
0
5
10
15
20
25
30
35
40
45
50
日齡

Characterization of the **vvIBDV** strains

- Pathotyping
  - Up to 100% mortality in SPF chickens
  - 50-60% mortality in layers, 25-30% in broilers
  - Break through higher levels of *maternally-derived antibodies* (MDA)

- Antigenicity 抗原性不變
  - Panel of neutralizing mAbs → no (major) antigenic drift

- Molecular characterisation
  - Sequencing of the **variable domain** of the **VP2** gene
  - RT-PCR of VP2 followed by sequencing and/or RFLP
  - Amino acid residues substitutions at positions 222Ala, 256Iso, 294Iso and 299Ser

    = Genetic fingerprints of vvIBDV

預防與控制

- 生物安全管理
- 統進統出、徹底消毒(碘)
- 疫苗
  - 種雞產蛋前注射死毒疫苗-->移行抗體保護小雞（被動免疫）
  - 適當時間注射活毒疫苗（主動免疫）
  - 不同雞場會有不同的防疫計畫

Prevent – Vaccine



投水口服



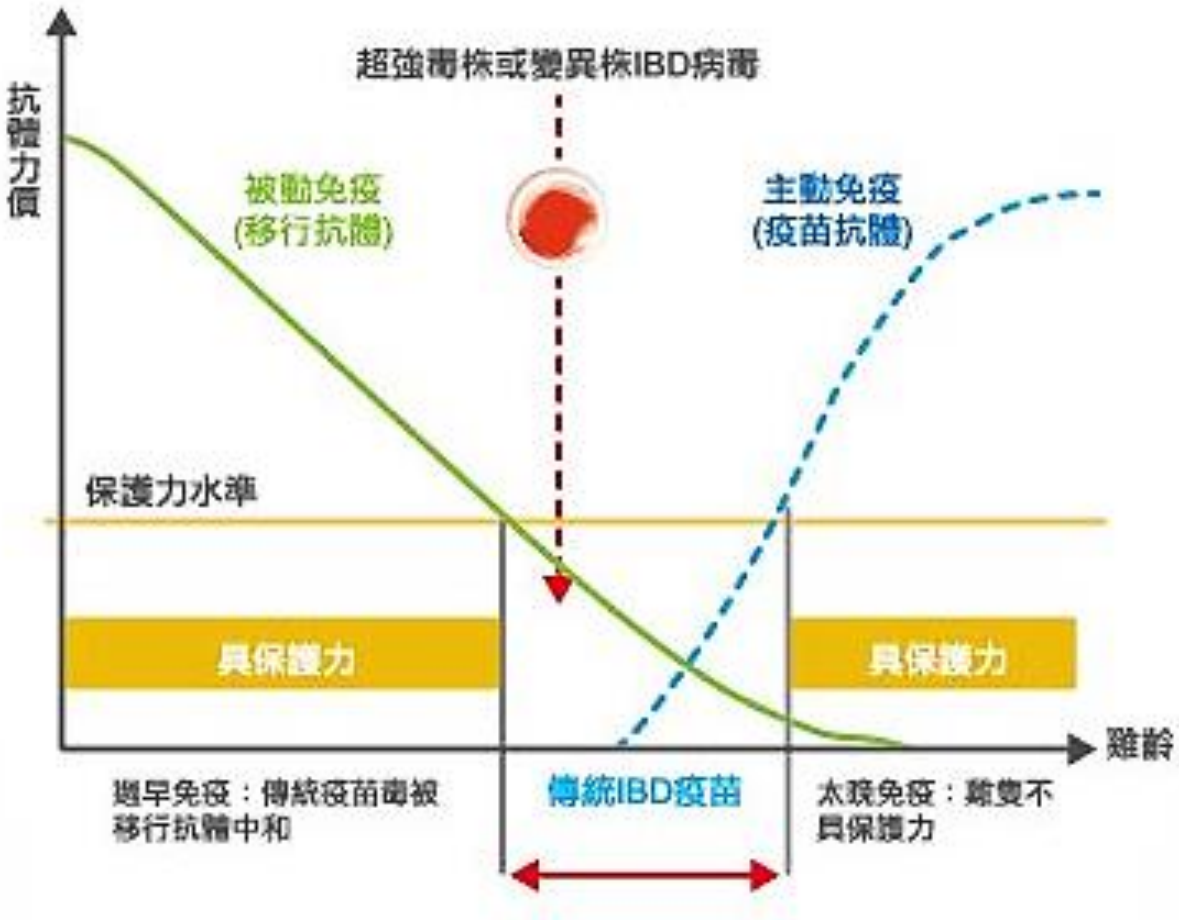
多價
肌肉注射



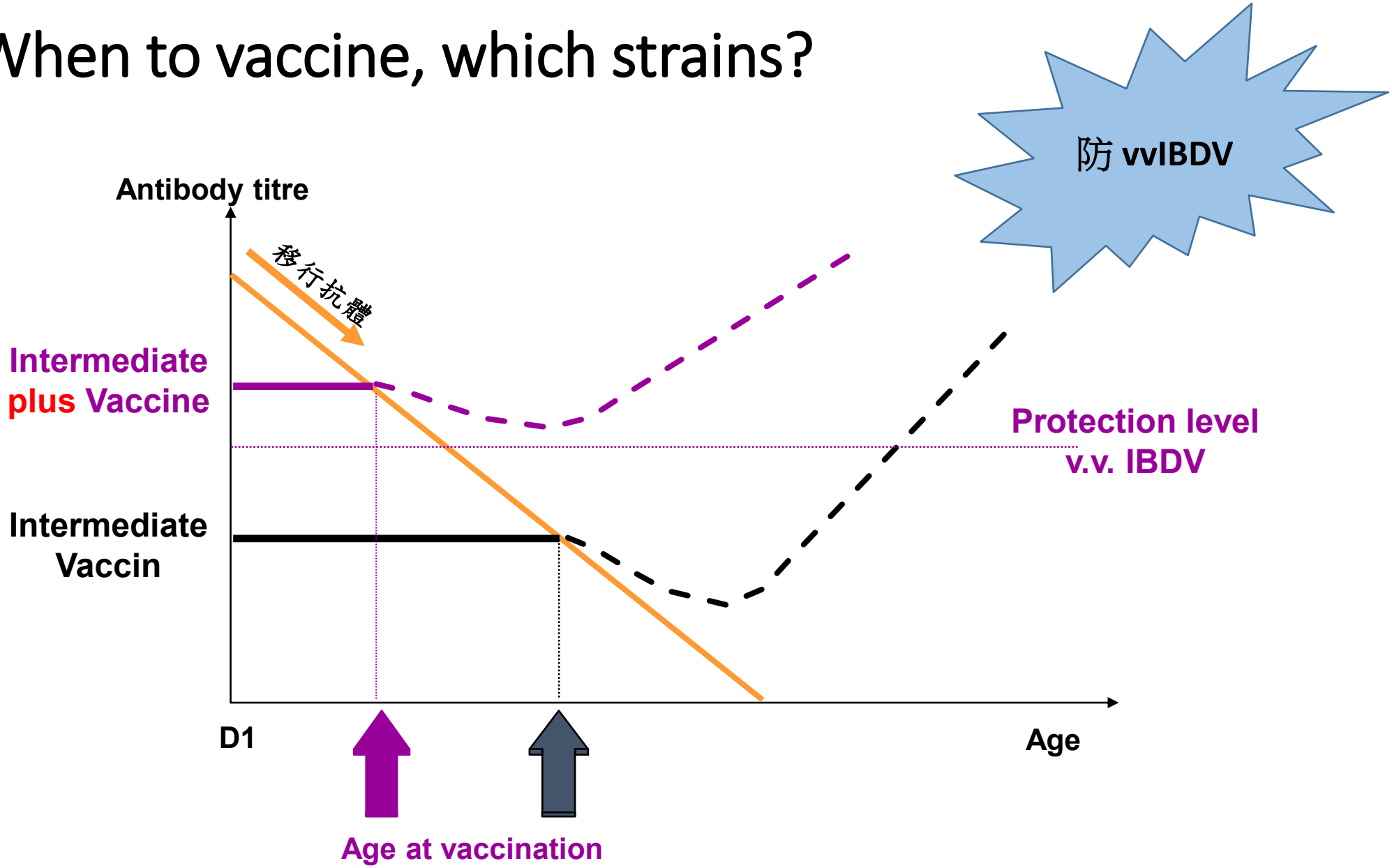
MD + IBDv 次單位

- Vaccine 的種類, 毒力, 劑型 非常多樣
- 施打前皆需考慮 “移行抗體力價”
- 須依據各場間 “各病毒的分佈, 以及過往IBDV的情形”作選擇

疫苗保護力 – 移行抗體空窗期

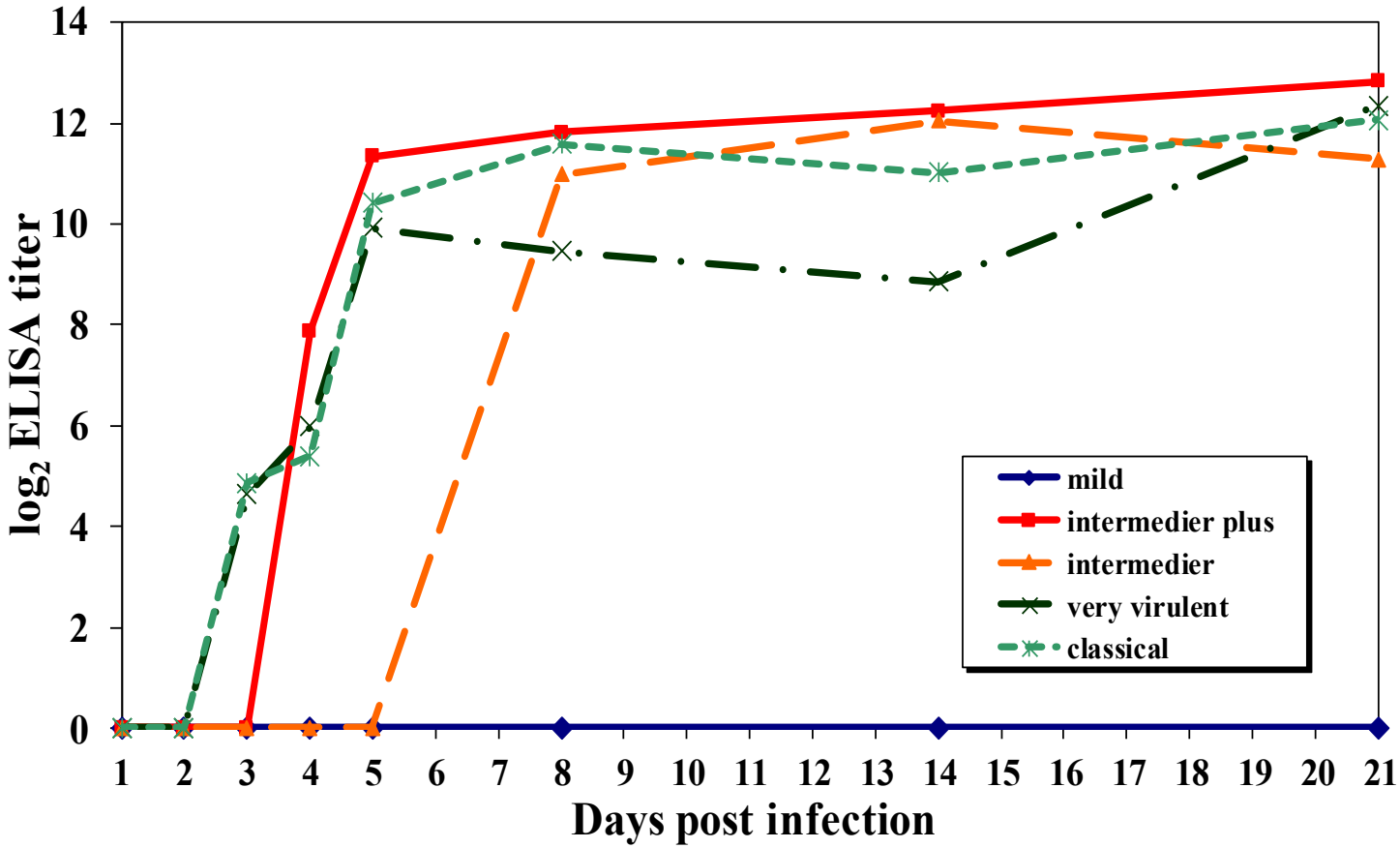


When to vaccine, which strains?



1. 中強毒 – 早 注射 及可不被移行抗體影響 – 早期 防範vvIBDV
2. 中間毒 – 晚 注射 才可不被移行抗體影響 – 較晚防範vvIBDV
3. Vaccine 的毒力不可超過移行抗體保護太多 → 會造成Bursa萎縮

Immunity induced by strains of different virulence



Vaccine strains: more **virulent** →

Faster induced immunity, but more **Damage** to bursa

→ what will happened to **other needed vaccine**?

IBDV 會造成 Immunosuppress

IBDV 感染後天數	免疫抑制程度 (以ND抗體力價觀察)
1	明顯
7	中度
14	無
21	無

(Faragher et al, 1974)

Conventional IBD Vaccines

- Mild IBD Vaccine - Easily eliminated by low levels of maternal antibody.
- Intermediate Vaccine - relatively safe to use *in ovo* or at hatch. Eliminated by low to moderate levels of maternal immunity.
- Intermediate Plus Vaccine - not safe to give *in ovo* unless modified in some manner. Eliminated by high level maternal immunity.

Immunity induced by strains of different virulence - CONCLUSIONS -

- Strains of higher virulence cause earlier bursa damage and earlier serological response
- Only strains that cause bursal damage induce immune response
- The tissue-culture adopted „mild” vaccine strain does not induced immune response in 4-week old SPF chicks (probably due to lack of receptors)

IBD: INNOVATIVE VACCINATION STRATEGY

Main issues around the protection against IBD:

- **Differences in level of MDA**
- **Variability in quality of vaccine application**
  - **Drinking water administration**
- **To adapt to the epidemiological context**

家禽重要疾病診斷研判圓桌會議結論

一、推薦之免疫計畫：

（一）危險區：4 日齡時施予弱毒飲水同時做 ND-IBD 油質死毒皮下注射。必要時於 18–21 日齡中間毒飲水補強。

（二）一般區：1 週內弱毒飲水，18–21 日齡中間毒飲水補強。（如果移行抗體力價差異大，比照危險區方法辦理。）

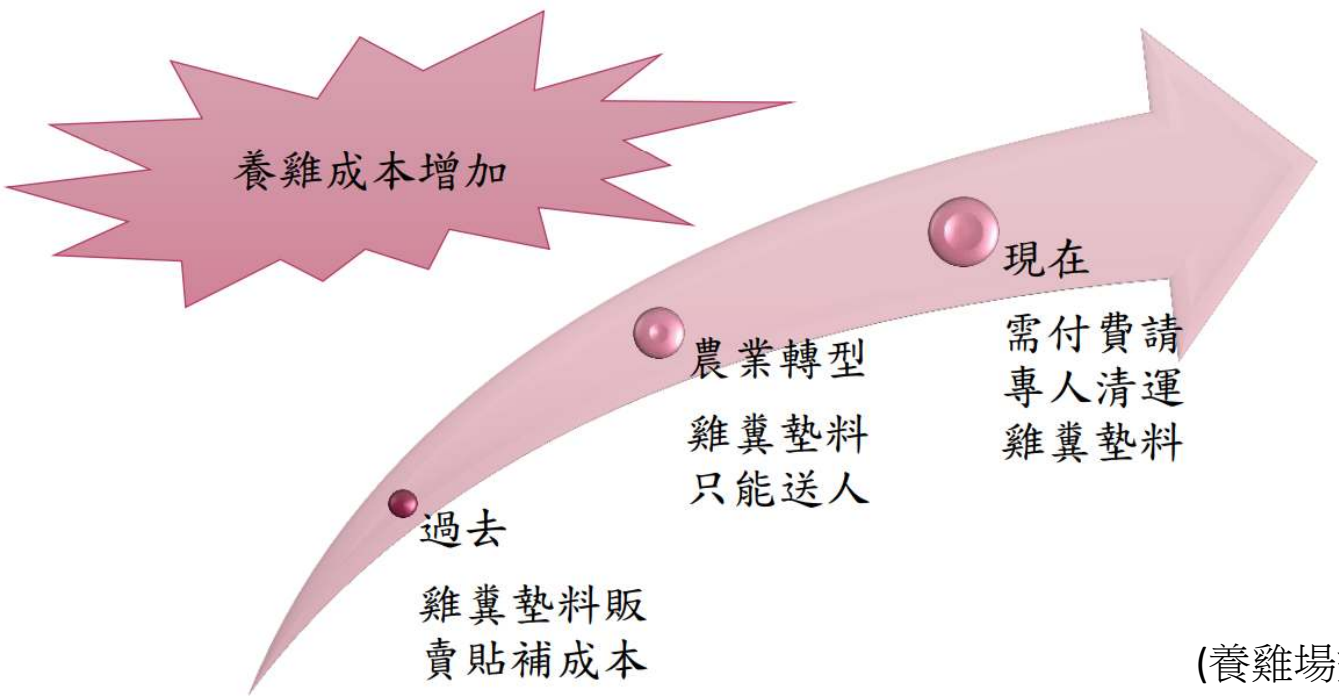
二、小雞應先經過弱毒疫苗基礎免疫後，才可使用中間毒疫苗，以免華氏囊遭受破壞而影響到所有疫苗之免疫效果。

IBD 疫苗接種宜先進行弱毒基礎免疫，若雛群未經血清學評估逕行使用中間偏強疫苗株疫苗，易造成華氏囊退行性變化，導致免疫功能下降

重複墊料的使用(109年後)

一、為何養雞場要重複使用墊料

在過去雞糞墊料可視為養雞業的副產品之一，可用於農作物施肥，但漸漸從可以販賣，到免費送人，直到現在甚至要付出費用請專人清運。



圖一 雞糞墊料之去化

(養雞場墊料重複使用示範手冊)

Summary

- IBD ≠ IB
- IBD = Infectious Bursal Disease = GUMBORO DISEASE
- 好發於小雞、急性、高傳染力的病毒性疾病
- 主要攻擊淋巴器官 (特別是bursa)
- 對環境抵抗力高
- 會造成嚴重的經濟損失
-